

Stellar Population Astrophysics (SPA) with the TNG

Beyond the iron peak: Galactic evolution of molybdenum, ruthenium, and zirconium through open clusters

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ABSTRACT

Context. Open clusters serve as key laboratories for studying the chemical evolution of the Milky Way. In particular, leveraging these coeval groups of stars to study elements beyond the iron peak provides insights into heavy-element nucleosynthesis, whose origins remain uncertain.

Aims. This work aims to determine the Mo, Ru, and Zr abundances for 81 stars across 30 open clusters. We compare these new chemical patterns with those of key reference elements (Y, Sr, and Eu), which were previously derived by our group. This comparison seeks to disentangle the distinct contributions from various nucleosynthetic processes.

Methods. Using the high-resolution spectra acquired within the Stellar Population Astrophysics programme using the HARPS-N echelle spectrograph at the Telescopio Nazionale Galileo, we estimated chemical abundances through spectral synthesis with TSFitPy under local thermodynamic equilibrium (LTE) conditions.

Results. This study provides the first Ru abundances for a large open cluster sample, alongside new measurements for Mo and Zr. We observe tight correlations between Mo, Ru, and other *s*-process elements (Sr, Y, and Zr), with slopes near unity, suggesting common enrichment timescales. Conversely, relations with the *r*-process element Eu show significant deviations, implying additional nucleosynthetic contributions. Crucially, chemical planes involving [Mo/Ru] and [s/Mo] ratios suggest systematic offsets when compared to theoretical *s*-process predictions.

Key words. stars: abundances – stars: evolution – Galaxy: abundances – Galaxy: disk – Galaxy: evolution – open clusters and associations: general

1. Introduction

Spanning the complete lifespan of the Galactic thin disc (Bossini et al. 2019), open clusters (OCs) provide a unique window into the chemo-dynamical evolution of the Milky Way (Cantat-Gaudin et al. 2018; Spina et al. 2022). Their importance in this field has been demonstrated across several recent works

(e.g. Carrera et al. 2019; Alvarez-Baena et al. 2024; Zhang et al. 2022; Carbajo-Hijarrubia et al. 2024; Dal Ponte et al. 2025, hereafter DP25). While Gaia (Gaia Collaboration 2016) provides unprecedented astrometric data for determining stellar distances and kinematics, its Radial Velocity Spectrometer (RVS) is constrained by a relatively modest spectral resolution and limited precision for detailed elemental abundances (see Carrera et al. 2022). Consequently, several ground-based spectroscopic surveys have been established to complement Gaia's reach. High- and medium-resolution projects such as the Gaia-ESO Survey (Gilmore et al. 2012; Randich et al. 2022),

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the Apache Point Observatory Galactic Evolution Experiment (APOGEE, [Majewski et al. 2017](#)), and the GALactic Archaeology with HERMES survey (GALAH, [De Silva et al. 2015](#)), alongside forthcoming projects such as the WHT Enhanced Area Velocity Explorer (WEAVE, [Jin et al. 2024](#)), The 4-metre Multi-Object Spectroscopic Telescope (4MOST, [de Jong 2019](#)), and the Multi-Object Optical and Near-infrared Spectrograph (MOONS, [Cirasuolo et al. 2014](#)), provide the necessary depth. These are further augmented by high-resolution, targeted surveys for OCs, including Open Clusters Chemical Abundances from Spanish Observatories (OCCASO, [Casamiquela et al. 2016](#)), One Star to Tag Them All (OSTTA, [Carrera et al. 2022](#)), and Stellar Population Astrophysics (SPA, [Origlia et al. 2019](#)). Together, these homogeneous datasets allow for a high-precision mapping of chemical abundance trends across the Galactic disc, revealing the Milky Way’s enrichment history ([Spina et al. 2022](#)).

Within this framework, neutron-capture elements serve as critical tracers of both nucleosynthetic processes and the broader chemical evolution of the Galaxy. While several investigations have successfully characterised neutron-capture species in OCs (e.g. [Magrini et al. 2018](#); [Sales-Silva et al. 2022](#); [Magrini et al. 2023](#); [DP25](#); [Bijavara Seshashayana et al. 2025](#)) including molybdenum (Mo, $Z = 42$; [Overbeek et al. 2016](#); [Mishenina et al. 2020](#); [Casamiquela et al. 2021](#); [Van der Swaelmen et al. 2023](#)), ruthenium (Ru, $Z = 44$) remains largely unobserved in statistically significant OC samples.

A joint analysis of Mo and Ru is essential, as their different fractional contributions from multiple nucleosynthetic processes (see Sect. 2) provide a powerful diagnostic of their origin when considered together. While previous studies have investigated Mo and Ru in the Galactic disc, halo, bulge, and field stars (see e.g. [Hansen et al. 2014](#); [Spite et al. 2018](#); [Mishenina et al. 2019, 2026](#); [Forsberg et al. 2022](#)), their behaviour in OCs remains largely unexplored. For this work, we exploited the high-resolution spectra of the SPA programme to derive, for the first time, Ru abundances for a large sample of OCs, alongside their Mo measurements, providing new insights into the chemical evolution of these neutron-capture elements in the Galactic disc. Additionally, we derived abundances for the s -process element zirconium (Zr, $Z = 40$) in our sample, and complement our analysis with the results obtained by [DP25](#), for the s -process elements strontium (Sr, $Z = 38$) and yttrium (Y, $Z = 39$), as well as the r -process element europium (Eu, $Z = 63$).

This paper is organised as follows. We present a general overview about the nucleosynthetic processes involved in the production of Mo and Ru in Sect. 2. In Sect. 3, we describe the observations and the selected sample. The determination of Mo, Ru, and Zr abundances are discussed in Sect. 4. Our results, together with a comparison to literature data and current models, are discussed in Sect. 5. Finally, Sect. 6 summarises our main findings and conclusions.

2. Production pathways of Mo and Ru

While the synthesis of lighter nuclei, up to and including iron (Fe, $Z = 26$), occurs through thermonuclear stellar fusion reactions, elements beyond Fe proceed through different pathways, among which neutron-capture is a major contributor ([Karakas & Lattanzio 2014](#)). This process unfolds in two stages: first, a neutron is captured by a seed nucleus, forming a heavier isotope; then, if the resulting nucleus is unstable, it undergoes a β^- -decay ($n \rightarrow p + e^- + \bar{\nu}_e$). The subsequent differentiation between the slow (s -) and rapid (r -) processes is given by com-

paring the rate of neutron-captures (which depends on the neutron flux density) and the β^- -decay timescale ([Burbidge et al. 1957](#); [Cowan et al. 2021](#)).

The s -process consists of three different components (see [Busso et al. 1999](#); [Bisterzo et al. 2014](#); [Mishenina et al. 2019](#), and references therein). The weak component that contributes to the s -isotopes from Fe to Sr, takes place in massive stars due to the $^{22}\text{Ne}(\alpha, n)^{25}\text{Mg}$ reaction, which is activated at the end of the convective He-burning core and in the subsequent convective C-burning shell. The main component contributing for the s -nuclei from Sr to Pb, is produced by low- to intermediate-mass asymptotic giant branch (AGB) stars, where neutrons are released by the $^{13}\text{C}(\alpha, n)^{16}\text{O}$ reaction in the radiative ^{13}C -pocket, formed right after the third dredge-up event and the $^{22}\text{Ne}(\alpha, n)^{25}\text{Mg}$ reaction that is mildly activated during the convective thermal pulses. The strong component is expected to provide at least 50% of ^{208}Pb , which is the most abundant nucleus produced in low metallicity AGB stars at $[\text{Fe}/\text{H}] \sim -1$ (e.g. [Gallino et al. 1998](#)).

Furthermore, the r -process is responsible for producing about half of the heavy elements, yet its dominant astrophysical site remains uncertain ([Cowan et al. 2021](#)). The two main candidate sources are core-collapse supernovae (CCSNe) and neutron star – neutron star (NS-NS) mergers ([Côté et al. 2018b](#)). The weak component of the r -process is associated with the neutrino-driven winds from CCSNe and electron-capture supernovae ([Wanajo et al. 2011](#); [Arcones & Thielemann 2013](#)). The main component is associated with the neutron-rich matter ejected by NS-NS mergers and neutron star – black hole mergers (see [Mishenina et al. 2019](#), and the references therein), as well as magneto-rotational supernovae (MR-SNe; [Reichert et al. 2021](#)).

An additional intermediate (i -) process was proposed by [Cowan & Rose \(1977\) where the neutrons are mainly produced by the reaction \$^{13}\text{C}\(\alpha, n\)^{16}\text{O}\$. As explained by \[Choplin et al. \\(2021\\)\]\(#\), the \$i\$ -process is triggered when hydrogen is mixed in a convective He-burning zone. Protons are transported down in the convective He-burning zone and are burned by the \$^{12}\text{C}\(p, \gamma\)^{13}\text{N}\$ reaction. The \$^{13}\text{N}\$ decays in \$\sim 10\$ minutes into \$^{13}\text{C}\$, being followed by the reaction \$^{13}\text{C}\(\alpha, n\)^{16}\text{O}\$ that operates mainly at the bottom of the convective helium-burning zone.](#)

Besides the neutron-capture, the proton-capture (or p -process) is responsible for the production of stable proton-rich nuclides that cannot be formed through neutron-capture processes (see [Arnould & Goriely 2003](#); [Rauscher et al. 2013](#), and references therein). The γ -process is the most established scenario for the production of the p -nuclei, which are synthesised via different photo-disintegration paths starting from heavier nuclei ([Pignatari et al. 2016](#)). It proceeds through reactions that can lead to the formation of p -nuclei in two ways: either by destroying pre-existing neutron-richer isotopes through sequences of (γ, n) reaction – which dominates the most stable nuclei – or by initiating reaction chains from heavier, unstable nuclei, where (γ, p) or (γ, α) reactions occur, followed by a series of β^- -decays that gradually produce proton-rich isotopes. The favoured scenario for the p -process is associated with the explosion of the Ne/O-rich layers of a massive star during its final core collapse as a Type II supernova (SNII). However, it fails to produce the required amounts of p -nuclei in several mass ranges ([Rauscher et al. 2013](#)). For this reason, other complementary mechanisms have been proposed, for example Type Ia Supernovae, He-accreting CO white dwarfs of sub-Chandrasekhar mass and proton-rich components of neutrino-driven winds (see [Mishenina et al. 2019](#), and all the references therein).

Leveraging the well-determined ages of OCs, studies have demonstrated that *s*-process abundances with respect to iron ($[s/Fe]$) increase in younger stellar populations – a trend first identified by D’Orazi et al. (2009) for barium (Ba, $Z=56$) and further explored later on (see Magrini et al. 2022, for a review). However, we notice that Ba has been proven to be very uncertain at a young age and lanthanum (La, $Z=57$) should always be preferred as an *s*-process tracer (D’Orazi et al. 2022).

Both Mo and Ru consist of seven stable isotopes each, produced through different nucleosynthetic channels, and measuring their abundances offers key insights into the chemical evolution of the Galaxy (Mishenina et al. 2026). The $^{92,94}\text{Mo}$ are *p*-only nuclei (Rauscher et al. 2013; Pignatari et al. 2016), while ^{96}Mo is a pure *s*-process isotope (Käppeler et al. 2011; Bisterzo et al. 2014) and ^{100}Mo is an *r*-only isotope (Travaglio et al. 2004). Additionally, the *i*-process could produce efficiently ^{95}Mo and ^{97}Mo (Côté et al. 2018a). Similarly, $^{96,98}\text{Ru}$ are *p*-only nuclei (Rauscher et al. 2013; Pignatari et al. 2016), while ^{100}Ru is a pure *s*-isotope (Bisterzo et al. 2014) and ^{104}Ru is an *r*-only isotope (Snedden et al. 2008; Prantzos et al. 2020). The different contributions for each isotope are presented in Table 1. While not considered in this classification, neutrino-driven ejecta from core-collapse supernovae have also been proposed to explain Mo and Ru observations in a number of metal-poor stars and for a possible impact on the GCE of these elements (e.g. Montes et al. 2007; Farouqi et al. 2010; Arcones & Montes 2011; Wanajo et al. 2011; Bliss et al. 2018; Psaltis et al. 2024). In this case, the relative contribution to the different isotopes may vary greatly depending on the properties of the components considered.

The even isotopes ($^{92,94,96,98,100}\text{Mo}$) have nuclear spin $I = 0$ and therefore produce no hyperfine splitting, while the odd isotopes ($^{95,97}\text{Mo}$), with spin $I = 1/2$, should exhibit such structure. However, Grace & More (1934) showed that the hyperfine components of the odd isotopes are extremely weak and become masked by the much stronger contributions of the even isotopes. In addition, small isotope shifts overlap with the expected hyperfine separations, so the observed Mo I lines appear only as broadened or asymmetric profiles in their lab spectra, preventing a clean measurement of hyperfine splitting. For this reason, the Mo isotopic ratios cannot be determined, but rather the Mo abundance as a whole, which enables the tracing of major contributions from various nucleosynthesis processes (Hansen et al. 2014; Forsberg et al. 2022). An analogous situation is expected for Ru (Murakawa 1955).

3. Observations and the cluster sample

The SPA Programme exploits the combination of the High Accuracy Radial velocity Planet Searcher for the Northern Hemisphere (HARPS-N) and GIANO-B echelle spectrographs at the 3.5-metre Telescopio Nazionale Galileo (TNG) at the Observatorio del Roque de los Muchachos on La Palma (Canary Islands, Spain). It aims at obtaining high-quality spectra of more than 500 stars in the Milky Way thin disc and associated star clusters at different Galactocentric distances, including the poorly explored inner disc.

The SPA project includes both young OCs, such as ASCC 123 (Frasca et al. 2019), Praesepe (D’Orazi et al. 2020), and Stock 2 (Alonso-Santiago et al. 2021), as well as older ones, such as Collinder 350, Gulliver 51, NGC 7044, and Ruprecht 171 (Casali et al. 2020). Additionally, SPA exploits the NIR instrument GIANO-B to study helium (Jian et al. 2024), phosphorus (Jian et al. 2026), fluorine (Bijavara Seshashayana et al. 2024),

and many other elements, to be compared with what we obtain from HARPS-N (Bijavara Seshashayana et al. 2025). In particular, DP25 estimated the non-local thermodynamic equilibrium (NLTE) atmospheric parameters and abundances of giant stars in 33 OCs, from the HARPS-N spectra. Their results will serve as the starting point for this study.

For this work, we restrict ourselves to the analysis of the HARPS-N optical spectra with a resolution of $R \approx 115\,000$ and a spectral coverage $\lambda \approx 3800\text{--}6900\text{ \AA}$ (Cosentino et al. 2012). The observations used for this study were obtained between July 2018 and April 2023.

Our sample consisted initially of 93 evolved stars on the red giant branch (RGB) and the red clump (RC) across 33 different OCs (see Fig. A.1) selected from high-probability OC members based on Cantat-Gaudin et al. (2018, 2020). These numbers differ from DP25 as we discarded the two stars in Theia 1214 (included as members of NGC 752 in DP25) since our analysis revealed large discrepancies in their chemical abundances both between the two stars and with respect to NGC 752. This result is in agreement with Ćurjurić et al. (2026), who performed a dedicated analysis of these two stars, concluding that they are most likely field stars. An additional constraint in effective temperature (T_{eff}) was applied: stars with T_{eff} below 4300 K were discarded as we were unable to properly estimate C and N for them as their resulting spectral fits showed an over-depression of the continuum that we believe to be related to the presence of TiO molecular bands. A more detailed discussion of this limitation will be presented in a dedicated paper (Alvarez-Baena, in prep.). After this cut, the final sample consists of 81 stars in 30 OCs as Gulliver 18, Gulliver 24 and NGC 7044 were entirely removed because all of their members fell below this threshold. The final number of stars in each OC, along with the general properties of the clusters, is presented in Table 2.

For this study, we used the stellar parameters and chemical abundances of Sr, Y and Eu derived by DP25. For the derivation of the stellar parameters, DP25 adopted the NLTE Optimization Tool Utilized for the derivation of atmospheric Stellar parameters (LOTUS) code¹ (Li & Ezzeddine 2023), which derives T_{eff} , surface gravity ($\log g$), metallicity ($[Fe/H]$) and microturbulence (v_{mic}) from Fe I and Fe II EWs using generalised curves of growth in LTE and NLTE. Parameters are obtained via global minimisation, with uncertainties estimated through a Markov chain Monte Carlo (MCMC) approach. The elemental abundances were obtained using NLTE synthesis (for details see DP25). Since macroturbulence (v_{mac}) values were not explicitly reported by DP25, we derived them by averaging the results from their successful Ti line fits. For two exceptions, UBC 169 star #2 and UBC 194 star #1, the v_{mac} values were instead calculated based on Ni line estimations. The adopted values are summarised in Table B.1.

4. Spectroscopic analysis

We used the open-source PYTHON wrapper TSFitPy² (Gerber et al. 2023) of the Turbospectrum 2020 code (Alvarez & Plez 1998; Plez 2012)³. This tool allows the computation of synthetic stellar spectra with NLTE for multiple chemical species at once by fitting the normalised synthetic spectra from Turbospectrum by χ^2 minimisation using the Nelder–Mead algorithm (Gerber et al. 2023). The tool was

¹ <https://github.com/Li-Yangyang/LOTUS>

² <https://github.com/TSFitPy-developers/TSFitPy>

³ https://github.com/bertrandplez/Turbospectrum_NLTE

Table 1. Solar abundances of Mo and Ru isotopes and their contribution by mass fraction as reported by [Sneden et al. \(2008\)](#).

Isotope	Solar Abundance (%) ^a	Processes	<i>s</i> -process ^h	<i>r</i> -process ^h	<i>p</i> -process
⁹² Mo	14.525	<i>p</i> -only ^{bcd}	–	–	1.000
⁹⁴ Mo	9.151	<i>p</i> -only ^{bcd}	–	–	1.000
⁹⁵ Mo	15.838	(<i>r</i> -, <i>s</i> -) ^{bhi} and <i>i</i> - ^g	0.470	0.530	–
⁹⁶ Mo	16.672	<i>s</i> -only ^{befhi}	1.000	–	–
⁹⁷ Mo	9.599	(<i>r</i> -, <i>s</i> -) ^{bhi} and <i>i</i> - ^g	0.642	0.358	–
⁹⁸ Mo	24.391	(<i>r</i> -, <i>s</i> -) ^{bhi}	0.847	0.153	–
¹⁰⁰ Mo	9.824	<i>r</i> -only ^{bh} or (<i>r</i> -, <i>s</i> -) ⁱ	–	1.000	–
⁹⁶ Ru	5.540	<i>p</i> -only ^{bcd}	–	–	1.000
⁹⁸ Ru	1.870	<i>p</i> -only ^{bcd}	–	–	1.000
⁹⁹ Ru	12.760	(<i>r</i> -, <i>s</i> -) ^{bi} or <i>s</i> -only ^h	1.000	–	–
¹⁰⁰ Ru	12.600	<i>s</i> -only ^{befhi}	1.000	–	–
¹⁰¹ Ru	17.060	(<i>r</i> -, <i>s</i> -) ^{bhi}	0.158	0.842	–
¹⁰² Ru	31.550	(<i>r</i> -, <i>s</i> -) ^{bhi}	0.444	0.556	–
¹⁰⁴ Ru	18.620	<i>r</i> -only ^{bhi}	–	1.000	–

Notes. ^(a) [Asplund et al. \(2009\)](#). ^(b) [Anders & Grevesse \(1989\)](#). ^(c) [Rauscher et al. \(2013\)](#). ^(d) [Pignatari et al. \(2016\)](#). ^(e) [Käppeler et al. \(2011\)](#). ^(f) [Bisterzo et al. \(2014\)](#). ^(g) [Côté et al. \(2018b\)](#). ^(h) [Sneden et al. \(2008\)](#). ⁽ⁱ⁾ [Prantzos et al. \(2020\)](#).

Table 2. Summary of the OC analysed in this work.

Cluster	Other names	SPA members ^a	N	Age (Gyr)	A _v (mag)	Distance (pc)	R _{GC} (pc)
Alessi 1	Casado-Alessi	4	4	1.45	0.08	689	8726
Alessi 161	UBC 3	1	1	0.13	0.98	1704	7223
Alessi-Teutsch 11	ASCC 112, Alessi 46	1	1	0.14	0.37	634	8335
Basel 11b	FSR 877	3	3	0.23	1.56	1793	10121
COIN-Gaia 30	–	1	1	0.26	1.25	767	8804
Collinder 350	–	2	1	0.59	0.52	371	8021
Collinder 463	–	2	2	0.11	0.79	849	8874
Gulliver 51	–	1	1	0.36	1.42	1536	9410
IC 4756	Collinder386, Melotte 210	12	12	1.29	0.29	506	7938
LP 1800	UBC 170, FoF 1800	3	3	0.33	1.25	1392	8809
NGC 2437	M 46, Melotte 75	5	5	0.30	0.73	1511	9345
NGC 2509	Melotte 81, Collinder 171	1	1	1.51	0.23	2495	9887
NGC 2548	M 48, Melotte 85	3	3	0.40	0.15	772	8857
NGC 2632	M 44, Praesepe	2	2	0.68	0.00	183	8479
NGC 2682	M 67	2	2	4.27	0.07	889	8964
NGC 6800	–	1	1	0.41	0.83	1012	7868
NGC 6991	–	5	5	1.55	0.20	577	8333
NGC 7086	Collinder 437	2	2	0.19	1.81	1677	8632
NGC 7209	Melotte 238, Collinder 444	2	1	0.43	0.53	1154	8525
NGC 752	Melotte 12, Theia 1214	5	3	1.17	0.07	483	8669
Ruprecht 171	–	7	4	2.75	0.68	1522	6895
Stock 2	–	8	8	0.40	0.50	399	8619
Tombaugh 5	–	3	3	0.19	2.07	1706	9768
UBC 141	–	1	1	2.09	0.51	1315	8176
UBC 169	–	2	1	0.30	0.26	1347	8749
UBC 194	–	1	1	0.23	0.48	1361	9394
UBC 577	Alessi 191	4	4	2.75	0.00	1122	7718
UBC 60	COIN-Gaia 11	1	1	0.79	1.25	669	8977
UPK 219	–	1	1	0.15	1.20	873	8735
UPK 84	UBC 131, Alessi 116	3	3	1.00	0.31	944	7980

Notes. For each cluster, we list the number of stars studied in this work (N) and in [DP25](#) (SPA members). The OC properties (Age, A_v, distance, R_{GC}) are from [Cantat-Gaudin et al. \(2020\)](#). ^(a) Stars studied in each OC by [DP25](#).

Table 3. Atomic data for the spectral lines analysed in this work.

Species	Wavelength in air (Å)	log gf (dex)	$\chi_{\text{exc}}^{\text{low}}$ (eV)
Mo I	5570.444	−0.337	1.335
Mo I	6030.644	−0.523	1.531
Ru I	4869.153	−0.830	0.928
Zr I	6127.440	−1.060	0.154
Zr I	6134.5500	−1.280	0.000

Notes. These data come from the line list of the *Gaia*-ESO Survey (Heiter et al. 2021) updated by Magg et al. (2022). The atomic parameters for Mo I, Ru I, and Zr I were originally reported by Whaling & Brault (1988), Wickliffe et al. (1994), and Biemont et al. (1981), respectively.

further optimised and had the addition of some extra features as described in Storm & Bergemann (2023), including the micro-turbulence optimisation as described in D’Orazi et al. (2024). The assumed solar abundances from Asplund et al. (2009) and Magg et al. (2022) are: $A(\text{Fe})_{\odot} = 7.50$, $A(\text{Sr})_{\odot} = 2.87$, $A(\text{Y})_{\odot} = 2.21$, $A(\text{Zr}) = 2.58$, $A(\text{Mo})_{\odot} = 1.88$, $A(\text{Ru})_{\odot} = 1.75$, and $A(\text{Eu})_{\odot} = 0.52$.

Our initial analysis focused on the Mo I lines at 5506 and 5533 Å (Mishenina et al. 2019, 2026) and the 6030 Å line (Forsberg et al. 2022). Following the *Gaia*-ESO line list classification (Heiter et al. 2021), the 6030 Å feature is categorised as a *Yes/Yes* line, indicating both a high-quality oscillator strength (log gf) value and an unblended profile. This classification system provides usage recommendations for spectral lines based on their performance in solar and Arcturus benchmark spectra. In contrast, the 5506 Å and 5533 Å lines are classified as *Yes/Undecided* and *Yes/No*, respectively. Upon inspecting our sample, we found the Mo I lines at 5506 and 5533 Å to be severely affected, even at very high resolution, by the presence of strong nearby Fe I features. This rendered any resulting fits unreliable; consequently, these two lines were excluded from our analysis. The 6030 Å line proved to be too weak for reliable measurement in the majority of our spectra. Relying solely on the 6030 Å feature would have reduced our effective sample size to approximately one-third of its original total. Due to the limitations found for the 6030 Å line, we tested several other lines, e.g. 5751.408 (*Yes/Yes*) and 5858.266 (*Yes/Undecided*). We found that a good alternative was the Mo I line at 5570.444 Å (*Yes/Undecided*) previously used by Roriz et al. (2021) for the Mo estimation in Ba stars and by Overbeek et al. (2016) for Mo determination in OCs. This line was successfully measured for the majority of our sample.

The Mo I lines at 5570 and 6030 Å are both affected by CN molecular blends, with the contribution being particularly significant for the 5570 Å feature. Although TSFitPy defaults to solar abundances, the giant stars in our sample are expected to exhibit nitrogen enhancement and carbon depletion (Karakas & Lattanzio 2014). To accurately model the CN blending, we independently determined C and N abundances for each star using the $^{12}\text{C}_2$ Swan (0,1) band with band head 5635 Å and the $^{12}\text{C}^{14}\text{N}$ molecular bands in the spectral range 6470–6490 Å. We adopt $[\text{O}/\text{Fe}] = 0.0$ dex because no telluric spectra were available. For more than 90% of our sample, the O I line at 6300.3040 Å is contaminated by telluric absorption features, a well-known limitation of this diagnostic (e.g. van der Plas et al. 2008; Jönsson et al. 2017). This assumption is supported by the

results of DP25, who found nearly solar metallicities for 33 OCs consistent with expectations for stellar populations in the Milky Way thin disc (Gaia Collaboration 2023a).

We adopted Mo abundances derived from the 5570 Å and 6030 Å lines only when they were well-defined – specifically, when they appeared as relatively strong ($\text{EW} \gtrsim 7$ mÅ), unsaturated, and sufficiently isolated features (see Fig. 1). Atomic data for these transitions were sourced from the *Gaia*-ESO line list (Heiter et al. 2021), incorporating updates from Magg et al. (2022) (see Table 3). Following Mishenina et al. (2019), these weak subordinate lines originate in deep atmospheric layers where LTE typically holds. Therefore, we performed our analysis under the LTE assumption, as departures are expected to be negligible and specific NLTE corrections are currently unavailable for these lines.

For the Ru determination, after testing several Ru I lines (e.g. 4757.764 Å, 4757.848 Å, 4584.443 Å, 5309.265 Å, 5636.237 Å), we adopted the Ru I line at 4869.153 Å (see Table 3 for the atomic parameters), classified as *Yes/Undecided* by Heiter et al. (2021). This line (see Fig. 1) was also employed by Roriz et al. (2021) in their study of Ba stars. There is a small blend with the Si I line at 4869.086 Å, which we assume as negligible for our Ru determination, as we obtained a median sensitivity of −0.01 dex when setting the $[\text{Si}/\text{Fe}] = 0.25$ dex. Here, the sensitivity refers to the change in the derived abundances resulting from a variation in the assumed input stellar parameter or abundance. The log gf and low-level excitation potential ($\chi_{\text{exc}}^{\text{low}}$) for this Si I line were determined in Pehlivan Rhodin et al. (2024).

Our Zr abundances were determined using the two Zr I lines at 6127.440 Å and 6134.550 Å (see Fig. 1), as previously done by Forsberg et al. (2019) and Mishenina et al. (2026). These Zr I lines are described by Velichko et al. (2010) as weak features with the presence of molecular blends. In general, Zr I is more prone to being affected by NLTE effects than Zr II, and Velichko et al. (2010) showed that over-ionisation can produce abundance corrections of about +0.30 dex for strong Zr I lines in RGB stars. In our analysis, we also explored several Zr II lines (e.g. 4050.220 Å, 4208.880 Å, 4257.941 Å, 4442.892 Å, 4496.862 Å, 5112.170 Å), as well as the Zr I lines at 4241.606 Å and 4687.705 Å used by Velichko et al. (2010), and the Zr I line at 6140.460 Å used by Forsberg et al. (2019). However, they failed to provide satisfactory fits for the observed spectra. Given that the Zr I lines considered here are weak, as explained by Mishenina et al. (2026), it remains uncertain whether NLTE corrections similar to those reported for stronger lines would apply in this case. The Zr I line at 6127.440 Å (*Yes/Yes*) shows a small blend with the Fe I line at 6127.3912 Å. On the other hand, the line at 6134.550 Å (*Yes/Yes*) shows a small feature mainly due to CN molecules. The atomic data for these two lines is presented in Table 3.

The elemental abundances were determined as the average of the values derived from individual spectral lines. As well as for the two Mo I lines, the estimated $[\text{C}/\text{Fe}]$ and $[\text{N}/\text{Fe}]$ were adopted as input parameters for the spectral synthesis of the Ru I and the Zr I. The associated uncertainties were computed as $\sigma/\sqrt{n}$, where σ represents the standard deviation among the measured lines and n is the number of lines used. In cases where only a single line was available, we adopted a representative uncertainty of 0.04 dex, corresponding to the median uncertainty of the sample. For Ru, which was measured using only one line, the uncertainty was instead estimated from the dispersion of measurements within each OC. For clusters with only one star, we again adopted the median value of 0.04 dex. At the cluster

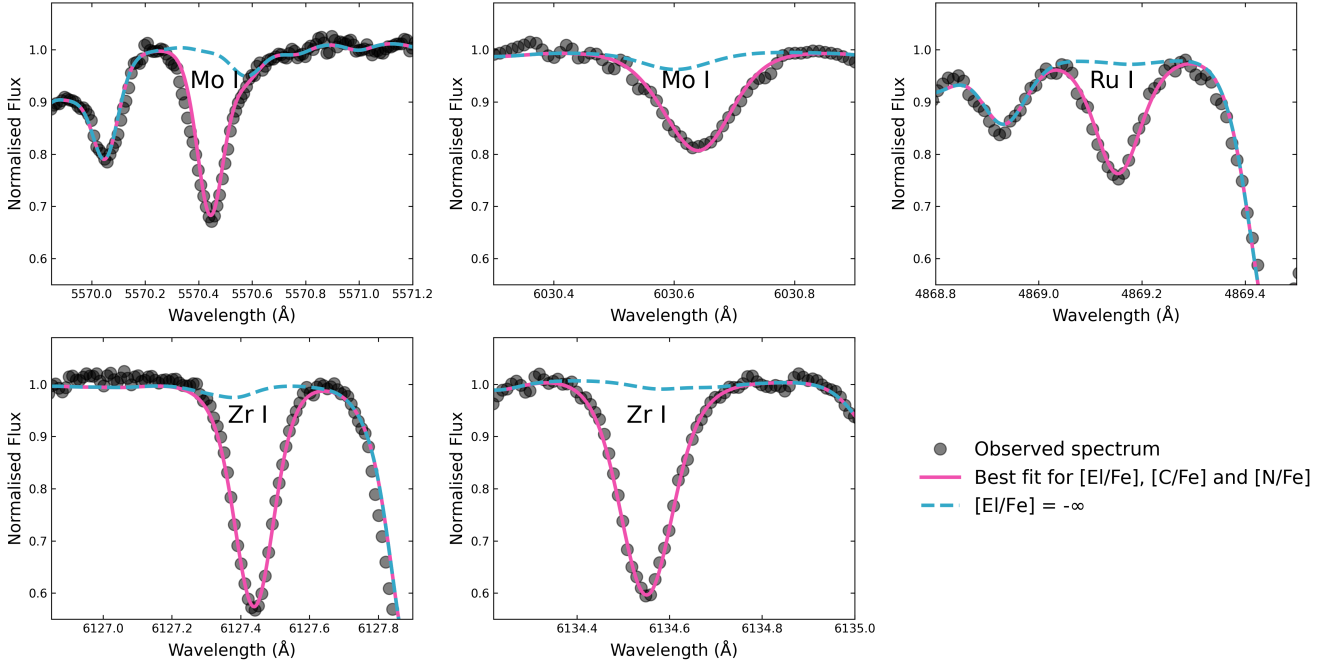


Fig. 1. Observed spectrum (black circles) for the star IC 4756 star #2. In each panel, the element under consideration (Mo, Ru or Zr) is indicated. The magenta solid line represents our best fit for each studied element using the estimated $[C/Fe]$ and $[N/Fe]$. The dashed teal line represents the synthetic spectrum computed without including that element.

level, abundances were computed as the mean of the stellar measurements within each OC, and the associated uncertainties were estimated as the standard deviation of the cluster divided by $\sqrt{N}$, where N is the number of stars with available measurements. For clusters with only one star, we adopted median uncertainties of 0.05 dex for $[Mo/Fe]$ and $[Zr/Fe]$, and 0.04 dex for $[Ru/Fe]$. The sensitivities to stellar parameters and C and N abundances are reported in Table D.

5. Results and discussion

In this section, we present the elemental abundances of Mo, Ru, and Zr derived in this work, compare our estimates with previous studies, and discuss the associated uncertainties. The derived abundances for Mo, Ru and Zr for individual stars are presented in Table C.1, while the corresponding mean abundances for each OC are listed in Table 4. In total, we obtained Mo abundances for 80 stars across 30 OCs, Ru abundances for 66 stars in 26 OCs, and Zr abundances for 81 stars in 30 OCs. All results were visually inspected to ensure the reliability and quality of the fit.

In Fig. 2 we present our results for $[Mo/Fe]$, $[Ru/Fe]$ and $[Zr/Fe]$ with respect to $[Fe/H]$. In this figure, we also present the results for thin and thick disc dwarfs (Mishenina et al. 2019), thin and thick disc giants (Forsberg et al. 2019, 2022) and thin disc giants (Mishenina et al. 2026, excluding HD 6833 due to its radial velocity (RV) measurement⁴). Additionally, we included the OCs from Overbeek et al. (2016), Mishenina et al. (2020), Casamiquela et al. (2021), Van der Swaelmen et al. (2023), excluding the OCs King 1 and NGC 2266 from Casamiquela et al. (2021), as their reported $[Mo/H]$ abundances are approximately 0.5 dex, corresponding

to deviations of about 10σ from the mean distribution, and are associated with comparatively large uncertainties.

The values from the literature were renormalised to our adopted solar scale (see Sect. 4). For the three planes in Fig. 2, there is an overall agreement between our results and those from the literature for the restricted metallicity range covered by our sample. In all three panels, the old OC Ruprecht 171 with $[Fe/H] \sim -0.06$ dex exhibits abundances lower than expected when compared with OCs of similar $[Fe/H]$. However, when compared with Casamiquela et al. (2021) (see Fig. 3) they obtained $[Mo/Fe] = -0.04 \pm 0.04$ dex and $[Zr/Fe] = -0.22 \pm 0.05$ dex which are consistent within the errors with our values of $[Mo/Fe] = -0.09 \pm 0.01$ dex and $[Zr/Fe] = -0.28 \pm 0.02$ dex.

Since the abundances from Casamiquela et al. (2021) were derived from different spectra using an independent methodology and linelist, this agreement supports the robustness of the low abundances, in particular the $[Zr/Fe]$, and makes it unlikely they are caused solely by observational limitations or systematic effects specific to our analysis. We estimated the guiding radius following Spina et al. (2021), using the galpy package and assuming the MW2014 potential for the Milky Way (Bovy 2015). We also examined the orbital parameters reported by Tarricq et al. (2021) for Ruprecht 171, but found no clear kinematical feature that could readily account for its peculiar chemical pattern. A more detailed kinematical analysis would be required to explore this possibility further, but it lies beyond the scope of this work.

Also, in all three panels, the intermediate-age OC NGC 6800 with $[Fe/H] \sim 0.11$ dex yields the highest abundances. For this cluster we obtained $[Mo/Fe] = 0.16 \pm 0.05$ dex, $[Ru/Fe] = 0.22 \pm 0.04$ dex and $[Zr/Fe] = 0.11 \pm 0.05$ dex. The three more metal-rich OCs (UBC 60 with $[Fe/H] \sim 0.23$ dex, NGC 2509 with $[Fe/H] \sim 0.21$ dex and NGC 2632 with $[Fe/H] \sim 0.16$ dex) seem to lie slightly off the distribution with the case of the $[Ru/Fe] = 0.15 \pm 0.04$ dex for UBC 60

⁴ HD 6833 has a reported RV of $-243.38 \text{ km s}^{-1}$ (Gaia Collaboration 2023b) which does not seem to be consistent with thin disc kinematics (see Anguiano et al. 2020).

Table 4. Mean [Mo/Fe], [Ru/Fe], and [Zr/Fe] obtained for each OC as well as the number of stars considered for the estimation of each element.

OC	[Mo/Fe] (dex)	$N_{\text{stars}}^{\text{Mo}}$	[Ru/Fe] (dex)	$N_{\text{stars}}^{\text{Ru}}$	[Zr/Fe] (dex)	$N_{\text{stars}}^{\text{Zr}}$
Alessi 1	0.08 ± 0.02	4	0.10 ± 0.02	4	-0.06 ± 0.01	4
Alessi 161	0.06 ± 0.05	1	0.13 ± 0.04	1	-0.04 ± 0.05	1
Alessi-Teutsch 11	0.02 ± 0.05	1	0.05 ± 0.04	1	-0.13 ± 0.05	1
Basel 11b	0.06 ± 0.03	3	–	0	0.01 ± 0.04	3
COIN-Gaia 30	0.05 ± 0.05	1	0.02 ± 0.04	1	0.00 ± 0.05	1
Collinder 350	0.12 ± 0.05	1	–	0	0.07 ± 0.05	1
Collinder 463	0.06 ± 0.02	2	0.14 ± 0.02	2	-0.09 ± 0.03	2
Gulliver 51	-0.04 ± 0.05	1	0.02 ± 0.04	1	-0.14 ± 0.05	1
IC 4756	0.11 ± 0.02	12	0.13 ± 0.02	11	0.00 ± 0.02	12
LP 1800	0.04 ± 0.02	3	0.05 ± 0.01	2	-0.06 ± 0.02	3
NGC 2437	0.09 ± 0.03	5	0.13 ± 0.01	4	-0.05 ± 0.04	5
NGC 2509	-0.09 ± 0.05	1	–	0	-0.14 ± 0.05	1
NGC 2548	0.07 ± 0.06	3	0.11 ± 0.10	2	-0.05 ± 0.06	3
NGC 2632	-0.03 ± 0.02	2	0.07 ± 0.02	2	-0.18 ± 0.04	2
NGC 2682	-0.01 ± 0.02	2	0.07 ± 0.01	2	-0.15 ± 0.01	2
NGC 6800	0.16 ± 0.05	1	0.22 ± 0.04	1	0.11 ± 0.05	1
NGC 6991	0.11 ± 0.03	5	0.14 ± 0.02	5	-0.01 ± 0.02	5
NGC 7086	0.02 ± 0.01	2	0.07 ± 0.01	2	-0.10 ± 0.01	2
NGC 7209	-0.01 ± 0.05	1	0.03 ± 0.04	1	-0.11 ± 0.05	1
NGC 752	0.01 ± 0.02	3	0.13 ± 0.03	3	-0.04 ± 0.01	3
Ruprecht 171	-0.09 ± 0.01	4	-0.06 ± 0.04	4	-0.28 ± 0.02	4
Stock 2	0.10 ± 0.03	8	0.13 ± 0.02	6	-0.01 ± 0.03	8
Tombaugh 5	0.11 ± 0.08	3	0.02 ± 0.04	1	-0.03 ± 0.06	3
UBC 141	0.00 ± 0.05	1	0.04 ± 0.04	1	-0.15 ± 0.05	1
UBC 169	0.08 ± 0.05	1	–	0	0.07 ± 0.05	1
UBC 194	0.02 ± 0.05	1	–	0	0.02 ± 0.05	1
UBC 577	0.13 ± 0.03	3	0.06 ± 0.04	3	0.07 ± 0.06	4
UBC 60	0.06 ± 0.05	1	0.15 ± 0.04	1	-0.04 ± 0.05	1
UPK 219	0.11 ± 0.05	1	0.11 ± 0.04	1	0.03 ± 0.05	1
UPK 84	0.14 ± 0.03	3	0.21 ± 0.02	3	0.05 ± 0.01	3

being the most noticeable. When comparing our results with Casamiquela et al. (2021) for NGC 2632, they found $[\text{Mo}/\text{Fe}] = 0.02 \pm 0.02$ dex and $[\text{Zr}/\text{Fe}] = -0.05 \pm 0.03$ dex with respect to our $[\text{Mo}/\text{Fe}] = -0.03 \pm 0.02$ dex and $[\text{Zr}/\text{Fe}] = -0.18 \pm 0.04$ dex. $[\text{Mo}/\text{Fe}]$ is consistent within 2σ while $[\text{Zr}/\text{Fe}]$ exhibits a deviation of about 3σ .

We compare in Fig. 3 the results for our OCs in common with Casamiquela et al. (2021), including NGC 2682 (M67) that is also in common with Overbeek et al. (2016) and Van der Swaelmen et al. (2023). We present the different estimates for $[\text{Mo}/\text{H}]$, $[\text{Zr}/\text{H}]$ and $[\text{Fe}/\text{H}]$ ⁵. When comparing with Casamiquela et al. (2021), we found our estimates to be systematically higher for $[\text{Mo}/\text{H}]$ and $[\text{Fe}/\text{H}]$, with the exception of Ruprecht 171, which is in good agreement with their reported value. Regarding $[\text{Zr}/\text{H}]$, we find no significant systematic offset between our results and those of Casamiquela et al. (2021). Specifically, the clusters Alessi 161, NGC 2632, NGC 6991, and Ruprecht 171 show good agreement within the uncertainties, with discrepancies generally remaining below 0.1 dex, except for NGC 2682. Notice that our $[\text{Mo}/\text{H}]$ and $[\text{Fe}/\text{H}]$ estimates for NGC 2682 are approximately 0.1 dex and 0.2 dex lower, respectively, than those reported by Overbeek et al. (2016). In contrast, our $[\text{Mo}/\text{H}]$ value for this cluster is in good agreement with Van der Swaelmen et al. (2023) within the uncertainties, though a difference of ~ 0.1 dex persists in $[\text{Fe}/\text{H}]$.

⁵ $[\text{El}/\text{Fe}] = [\text{El}/\text{H}] - [\text{Fe}/\text{H}]$.

To investigate the nucleosynthetic origin of Mo and Ru, we compare their abundances with those of elements whose nucleosynthesis production channels are well established (Hansen et al. 2014; Mishenina et al. 2019). For this purpose, we analyse the slopes defined by the absolute abundances $A(\text{El})$ ⁶ between the elements measured in this work (Mo, Ru, and Zr) and those reported by DP25, in particular, the neutron-capture elements Sr, Y (predominantly produced by the *s*-process) and Eu (mainly produced by the *r*-process). Since the comparison is performed using absolute abundances, it is not affected by the systematic offsets introduced by differences in the adopted solar abundance scale or by the use of Fe as reference element in abundance ratios.

For obtaining the slopes, we follow the procedure described by Nunnari et al. (2026). Using a Bayesian approach to infer model parameters (slope m and intercept b following the equation $y = mx + b$), along with their uncertainties. For a dataset D and parameters θ , the posterior probability is given by

$$p(\theta | D) \propto \mathcal{L}(D | \theta)\pi(\theta)$$

where $\pi(\theta)$ represents the prior distribution and $\mathcal{L}$ is the likelihood. In practice, we used the log-posterior

$$\log p(\theta | D) = \log \mathcal{L}(D | \theta) + \log \pi(\theta) + \text{const}$$

⁶ $[\text{El}/\text{H}] = A(\text{El}) - A(\text{El})_{\odot} = \log\left(\frac{n_{\text{El}}}{n_{\text{H}}}\right)_{*} - \log\left(\frac{n_{\text{El}}}{n_{\text{H}}}\right)_{\odot}$.

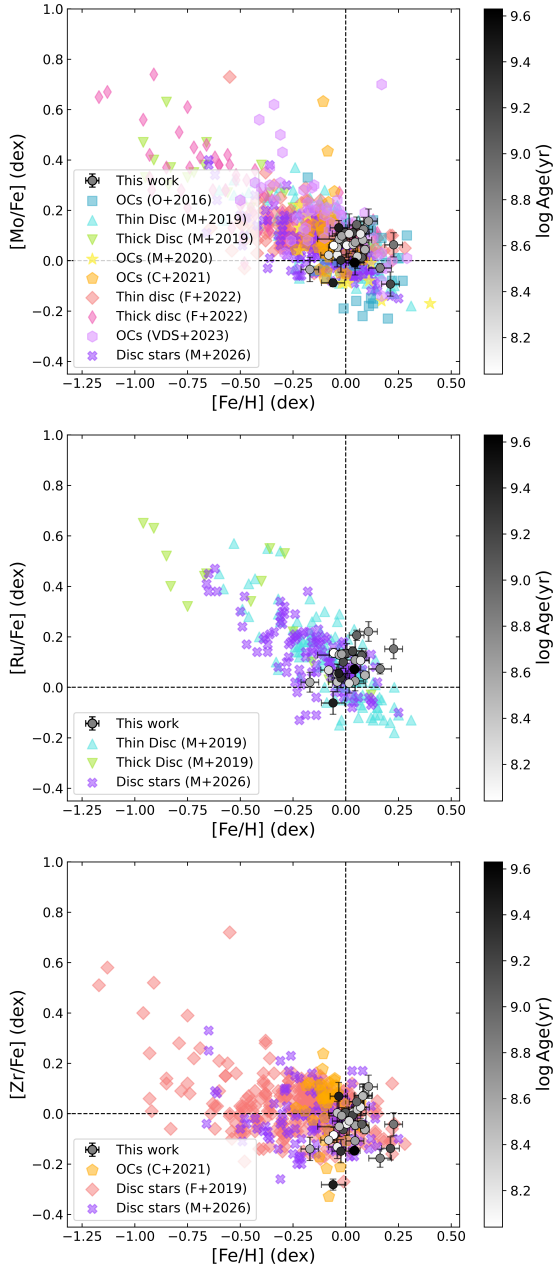


Fig. 2. [Mo/Fe] (upper panel), [Ru/Fe] (middle panel), and [Zr/Fe] (bottom panel) with respect to [Fe/H]. For each OC, our results are presented as circles, colour-coded by its age. The teal squares represent the OCs from [Overbeek et al. \(2016, O+2016\)](#), the upward cyan triangle and the downward green triangle represent the thin and thick disc stars from [Mishenina et al. \(2019, M+2019\)](#). The yellow stars are the OCs from [Mishenina et al. \(2020, M+2020\)](#), while the orange pentagons are the OCs from [Casamiquela et al. \(2021, C+2021\)](#). The red rotated squares and the pink diamonds are the thin and thick disc stars from [Forsberg et al. \(2022, F+2022\)](#) respectively in the case of [Mo/Fe], while for [Zr/Fe], the red rotated squares represent all the disc stars from [Forsberg et al. \(2019, F+2019\)](#). The light purple hexagons are the OCs from [Van der Swaelmen et al. \(2023, VDS+2023\)](#) and the dark purple crosses are the thin disc giants from [Mishenina et al. \(2026, M+2026\)](#). The dashed black lines that go through [0, 0] indicate the solar value.

Posterior sampling is performed using the affine-invariant Markov Chain Monte Carlo (MCMC) algorithm from the `emcee`⁷ library ([Foreman-Mackey et al. 2013](#)), which allows us

⁷ <https://github.com/dfm/emcee>

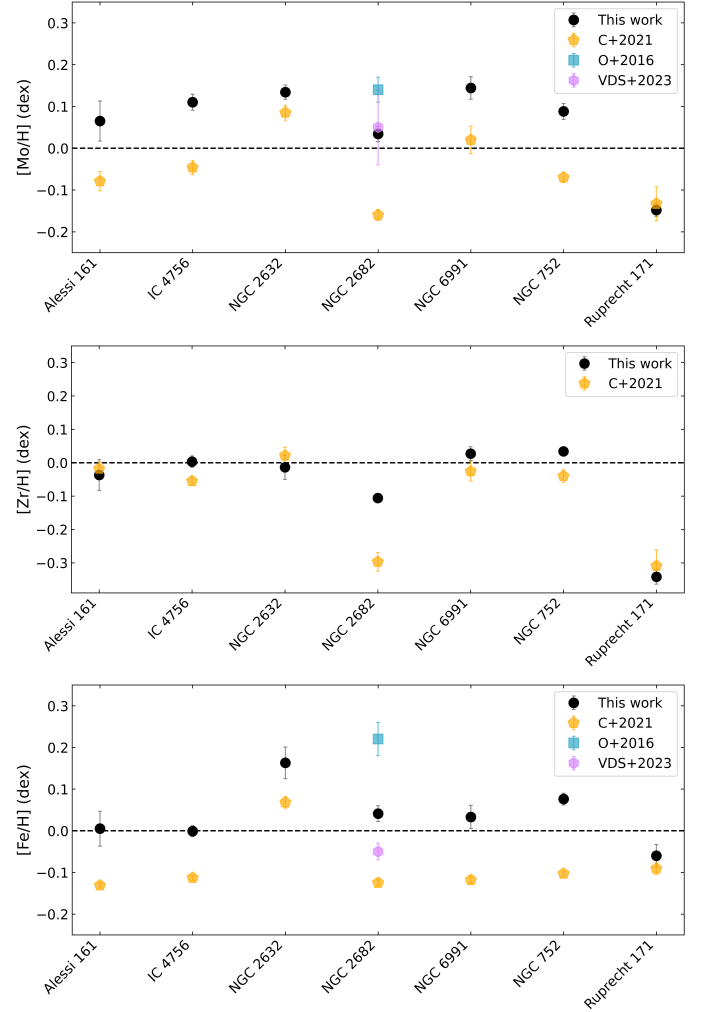


Fig. 3. Comparison of the values of [Mo/H] (upper panel), [Zr/H] (middle panel), and [Fe/H] (bottom panel) derived in this study (black circles) for the OCs in common with [Casamiquela et al. \(2021, C+2021\)](#), [Overbeek et al. \(2016, O+2016\)](#), and [Van der Swaelmen et al. \(2023, VDS+2023\)](#), represented by orange pentagons, teal squares, and light purple hexagons, respectively, with their corresponding errors. The [Fe/H] values labelled as ‘This work’ are adopted from [DP25](#).

to obtain maximum a posteriori (MAP) estimates and credible intervals corresponding to the 68% region of the marginal distributions.

We specifically use mixture models ([Foreman-Mackey et al. 2014](#)), where the likelihood of each data point is expressed as a weighted combination of a foreground (signal) and a background (outlier) component. Let Q denote the probability that a data point arises from the background, called the mixing fraction. Then, for a datum (x_i, y_i) with uncertainties $(\sigma_{x,i}, \sigma_{y,i})$, the individual likelihood is

$$\mathcal{L}_i = (1 - Q)\mathcal{L}_{fg}(x_i, y_i, \sigma_{x,i}, \sigma_{y,i} | \theta_{fg}) + Q\mathcal{L}_{bg}(x_i, y_i, \sigma_{x,i}, \sigma_{y,i} | \theta_{bg}),$$

and the total log-likelihood is

$$\log \mathcal{L} = \sum_i \log \mathcal{L}_i.$$

Hence, the mixing fraction Q controls the influence of the background model, making the inference less sensitive to outliers. In Gaussian implementations, each component is typically modelled as a normal distribution with its own mean and variance, and may also include an additional ‘jitter’ term, added in

quadrature, to account for potential underestimation of errors. For numerical stability, the evaluation of $\log \mathcal{L}$ should employ log-sum-exp techniques, which prevent underflow when one component contributes much less than the other.

Convergence of the MCMC chains was assessed using estimates of the autocorrelation time and by visual inspection of the trace plots. To quantify the quality of the fits, we report the mean squared error and the Akaike Information Criterion (AIC). The AIC provides a measure for model comparison that balances goodness-of-fit against model complexity. It is defined as:

$$\text{AIC} = 2k - 2 \log \mathcal{L}_{\max}$$

Where k is the number of model parameters and $\mathcal{L}_{\max}$ is the maximised likelihood. Models with lower AIC values are generally preferred, as they indicate a better compromise between fit quality and model simplicity.

The use of the mixture models instead of a conventional linear least-squares fitting was primarily motivated by the fact that the latter is only appropriate when the uncertainties are negligible along one axis and can be described by Gaussian distributions of known variance along the other (Hogg et al. 2010). These requirements do not hold for our observational data. Additionally, as it is also explained by Hogg et al. (2010), the mixture model provides a more appropriate treatment of outliers. Manually excluding these points or applying iterative sigma clipping would require subjective choices concerning the rejection threshold and the stage at which the rejection is performed. In contrast, the mixture model provides a probabilistic treatment of the full dataset.

Then, using this approach, we derive the slopes for our OCs as shown in Fig. 4, and compare them with the slopes reported by Mishenina et al. (2019). Since no slopes were provided for the thin disc stars in Forsberg et al. (2019, 2022) and Mishenina et al. (2026), nor for the OCs in Overbeek et al. (2016), Casamiquela et al. (2021) and Van der Swaelmen et al. (2023), we applied the same method to these samples. The resulting slopes for the available elements are reported in Table 5.

When comparing A(Ru) and A(Mo), we found a strong correlation between Mo and Ru, consistent with the fact that both elements have isotopes produced through a combination of p -, r -, and s -process nucleosynthesis. Our results are in good agreement with Mishenina et al. (2026) for thin disc giants, while showing a discrepancy with the slope for the thin disc dwarfs reported in Mishenina et al. (2019). Offsets between dwarfs and giants have been previously reported and may arise as a consequence of the analysis itself (see D’Orazi et al. 2009, for a discussion on this issue).

For the slopes between A(Eu) and A(Mo), we find a slope of ~ 0.5 , indicating that the r -process alone is insufficient to account for the observed Mo abundances, as expected from the isotopic contributions summarised in Table 1. Our results are consistent with those reported for thin disc stars by Mishenina et al. (2019) and Forsberg et al. (2019, 2022), as well as for the OCs by Van der Swaelmen et al. (2023). However, while the A(Mo) range derived by Overbeek et al. (2016) agrees with ours, we find a discrepancy in A(Eu). Investigating the origin of this difference is beyond the scope of this work, but it may arise from differences in the Eu abundance determination in Overbeek et al. (2016). Consistently, we derived a slope for A(Eu) versus A(Ru) of ~ 0.39 , also in agreement with those of Mishenina et al. (2019), while for A(Eu) versus A(Zr) we obtain ~ 0.42 . For the s -process elements (Sr, Y, Zr) we find slopes near unity when compared with Mo and Ru and among themselves, implying

Table 5. Linear-fit parameters for $y = mx + b$, obtained using the fitting procedure described in Sect. 5.

Relation A(El_y -axis) vs. A(El_x -axis)	m	b (dex)
This work		
A(Ru) vs. A(Mo)	0.97 ± 0.07	-0.04 ± 0.13
A(Eu) vs. A(Mo)	0.50 ± 0.10	-0.33 ± 0.19
A(Eu) vs. A(Ru)	0.39 ± 0.10	-0.08 ± 0.19
A(Eu) vs. A(Zr)	0.41 ± 0.09	-0.40 ± 0.22
A(Sr) vs. A(Mo)	0.89 ± 0.11	0.99 ± 0.21
A(Sr) vs. A(Ru)	0.79 ± 0.10	1.25 ± 0.20
A(Sr) vs. A(Zr)	0.76 ± 0.08	0.80 ± 0.21
A(Y) vs. A(Mo)	1.01 ± 0.09	0.21 ± 0.18
A(Y) vs. A(Ru)	0.91 ± 0.07	0.48 ± 0.14
A(Y) vs. A(Zr)	0.89 ± 0.07	-0.07 ± 0.17
A(Zr) vs. A(Mo)	1.17 ± 0.06	0.26 ± 0.12
A(Zr) vs. A(Ru)	1.05 ± 0.07	0.57 ± 0.13
Overbeek et al. (2016)		
A(Eu) vs. A(Mo)	0.11 ± 0.17	0.12 ± 0.35
Casamiquela et al. (2021)		
A(Y) vs. A(Mo)	1.12 ± 0.03	0.22 ± 0.06
A(Zr) vs. A(Mo)	1.23 ± 0.25	0.23 ± 0.51
Forsberg et al. (2019, 2022)		
A(Eu) vs. A(Mo)	0.66 ± 0.04	-0.59 ± 0.07
A(Zr) vs. A(Mo)	0.93 ± 0.04	0.71 ± 0.07
Van der Swaelmen et al. (2023)		
A(Eu) vs. A(Mo)	0.45 ± 0.06	-0.32 ± 0.12
Mishenina et al. (2026)		
A(Ru) vs. A(Mo)	1.00 ± 0.07	-0.05 ± 0.13
A(Sr) vs. A(Mo)	1.61 ± 0.07	-0.13 ± 0.12
A(Sr) vs. A(Ru)	1.28 ± 0.06	0.60 ± 0.11
A(Zr) vs. A(Mo)	1.15 ± 0.15	0.35 ± 0.12
A(Zr) vs. A(Ru)	1.31 ± 0.14	0.15 ± 0.24

Notes. The reported uncertainty is the average of the upper and lower uncertainties.

similar enrichment timescales. These results agree with findings for thin-disc stars by Mishenina et al. (2019, 2026) and Forsberg et al. (2019, 2022), as well as with the OCs reported by Casamiquela et al. (2021), i.e. OCs show slopes consistent with field stars. The offsets in fitted parameters between studies likely arise from differences in spectroscopic analysis, e.g. line selection, atomic data, and analysis code (see Jofré et al. 2017, 2019, for a more complete discussion on this topic). Offsets relative to Mishenina et al. (2019, 2026) and Forsberg et al. (2019, 2022) may additionally reflect that OCs are, on average, younger than field stars at the same metallicity.

We investigate chemical planes to gain insight into the stellar sources and to place constraints on their nucleosynthetic origin. For this purpose, we follow the approach by Mishenina et al. (2026), who analysed abundance patterns independently of metallicity by comparing $[\text{Zr}/\text{Mo}]$ vs $[\text{Ru}/\text{Mo}]$ ratios. We replicated this chemical plane in Fig. 5 and extended the analysis by additionally exploring the $[\text{Sr}/\text{Mo}]$ and $[\text{Y}/\text{Mo}]$ ratios versus $[\text{Mo}/\text{Ru}]$ for comparison. Consistent with Mishenina et al. (2026), none of the stars exhibits lower $[\text{Ru}/\text{Mo}]$ or higher $[\text{Zr}/\text{Mo}]$ ratios than those predicted for the s -process by Galactic Chemical Evolution (GCE) models (Bisterzo et al. 2014). Across all three panels in Fig. 5, we find a systematic offset from predicted s -process ratios. This suggests that the s -process alone cannot account for the observed $[\text{Ru}/\text{Mo}]$ values, as

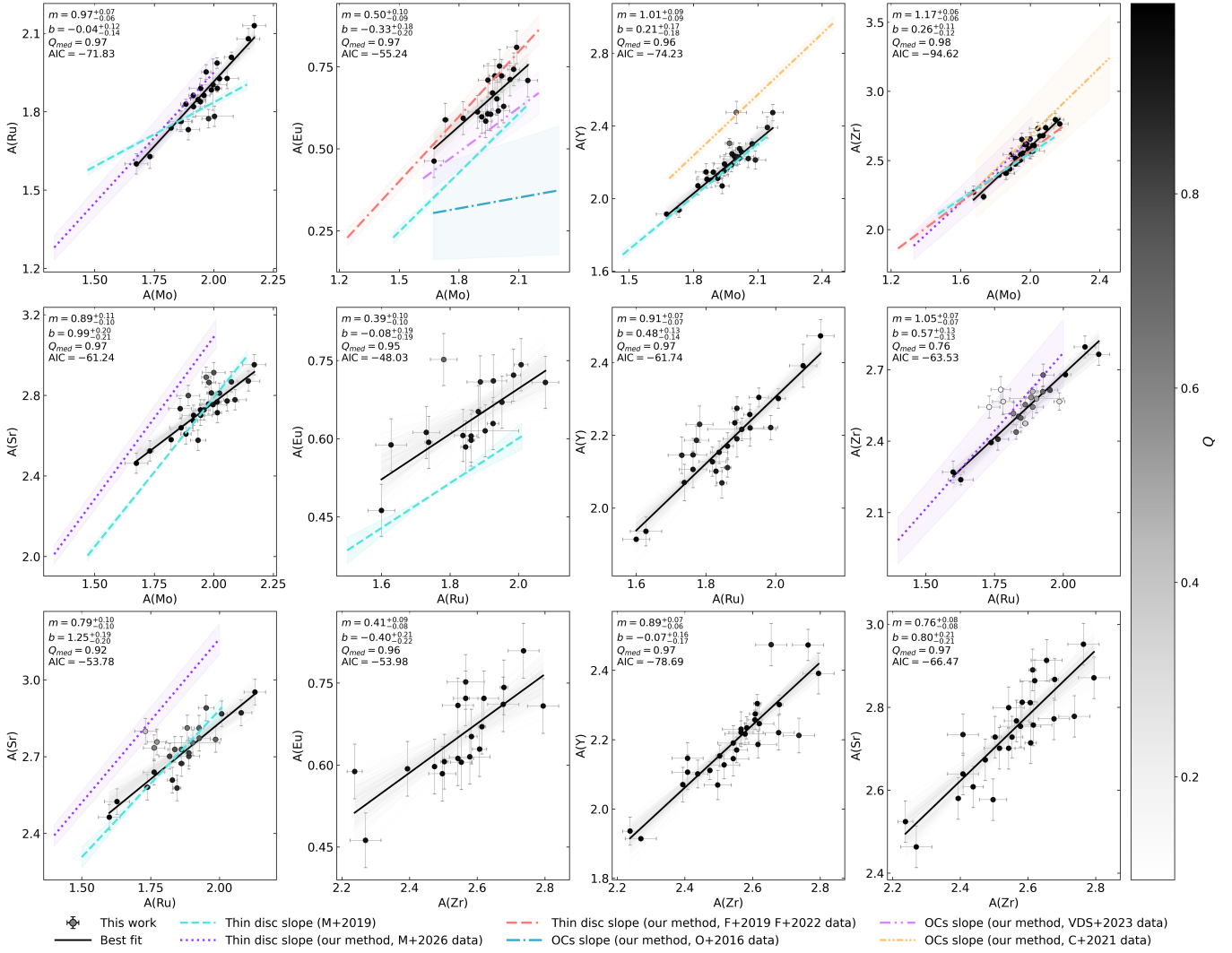


Fig. 4. Linear trends between the absolute abundances $A(\text{El}_y\text{-axis})$ vs. $A(\text{El}_x\text{-axis})$ of the measured elements. The OCs of this work are shown as circles, colour-coded in greyscale according to their quality factor Q . The black solid line represents the best-fit relation derived using our method described in Sect. 5. For each panel, the fitted slope m and intercept b , along with their uncertainties, as well as the median quality factor Q_{med} and the AIC, are reported. For comparison, the slopes reported in Mishenina et al. (2019) are plotted as cyan dashed lines. Additionally, using our method, we fitted the data from Mishenina et al. (2026) shown as a dotted dark purple line and the thin disc stars from Forsberg et al. (2019, 2022) as red dashdashdotted lines. The OCs from Overbeek et al. (2016), Van der Swaelmen et al. (2023) and Casamiquela et al. (2016) as teal dash-dotted lines, light purple dashdotted lines and yellow densely dashdotted lines, respectively. The best-fit parameters for this work and for the literature are reported in Table 5.

expected from GCE calculations for the MW disc and the Sun (Bisterzo et al. 2014; Prantzos et al. 2020). We exclude NLTE effects as the cause of this offset: our Sr and Y abundances are NLTE-corrected, and for Zr, Mo, and Ru, we utilised weak lines, minimising NLTE impacts at the target metallicities.

The r -process contribution is usually estimated as the residual remaining after subtracting the s -process fraction ($r = 1 - s$). Although this approach provides a useful reference, it implicitly assigns all yields not produced by the s -process to the r -process, neglecting the contributions from other nucleosynthetic channels, such as the p - and i -processes, which have been proposed as production channels for Mo and Ru (see Sect. 2). For this reason, and following Mishenina et al. (2026), we adopt the r -II star CS 22892-052 (Snedden et al. 2003) as a representative of the r -process production in the Zr–Ru region, as it exhibits an abundance pattern consistent with the solar r -process residual for elements heavier than Ba (Snedden et al. 2008). While this star

matches the solar r -process pattern for heavy elements (beyond Ba), this agreement breaks down for lighter elements between Sr and Ba (see Mishenina et al. 2026, and references therein).

Such a discrepancy led to the introduction of the LEPP (Lighter Element Primary Process, Travaglio et al. 2004; Montes et al. 2007), an additional nucleosynthesis component required to reproduce solar abundances originally associated with the abundances observed in several metal-poor stars. The predicted LEPP component is also shown in Fig. 5, however, its contribution seems insufficient to explain the offset.

In the figure, we also include the different conditions of the neutrino-driven ejecta with varying electron fraction (Y_e) from Psaltis et al. (2024), noting that nucleosynthesis can proceed through either the weak r - ($Y_e < 0.5$) or the νp -process ($Y_e > 0.5$). Finally, we report the typical range of abundance ratios obtained under i -process conditions, calculated using the same trajectory reproducing i -process neutron density conditions

used by Mishenina et al. (2026), for neutron exposures yielding an efficient production of the Zr–Ru mass region without Ba production. There are no Ba abundances for stars in our sample, but it is still a matter of debate the presence of enhanced Ba in young OCs, whether due to an *i*-process contribution or as an observational artefact (e.g. Mishenina et al. 2015; Baratella et al. 2021). However, for higher neutron exposures and higher Ba production, the *i*-process ratios are also shown in Fig. 5 and are concentrated around the highest [Ru/Mo] values reported.

From the comparison between our observations and the different nucleosynthesis components considered, we cannot clearly distinguish whether the location of our thin disc stars (OCs and field stars) might be reproduced by a simple mixture of *s*- and *r*-processes, or whether additional components would be required. This confirms the conclusion of Mishenina et al. (2026) for field stars concerning the observed Mo, Ru, and Zr abundance patterns. After the unexpected anomalies discovered in the Ba region (D’Orazi et al. 2009; Mishenina et al. 2015; Baratella et al. 2021), we cannot assume a perfect match between the signatures found in the field stars of the Galactic disc and OCs. Our results are based on a large, homogeneously analysed sample of OCs, which offer several advantages over field-star samples. Their members share a common age, kinematics, Galactic origin, and initial chemical composition, allowing properties such as age, distance, and chemical composition to be determined more accurately than for many other Galactic tracers (Friel 2013), allowing us to further explore the chemical signatures in the Sr–Ru region.

6. Summary and conclusions

This work is based on high-resolution spectra from the SPA programme obtained with the HARPS-N echelle spectrograph at the TNG. Chemical abundances of Mo, Ru and Zr were derived through spectral synthesis using TSFItPy for 81 stars in 30 OCs. In particular, we provide the first Ru abundance measurements for a large sample of OCs.

To investigate the nucleosynthetic origins of Mo and Ru, we examine the relationships between their abundances and those of other neutron-capture elements (Sr, Y, Zr, and Eu). We find that Mo and Ru are tightly correlated, exhibiting slopes near unity when compared to each other and to *s*-process tracers (Sr, Y, and Zr). These results indicate similar enrichment timescales and are consistent with previous studies of thin-disc stars and OCs. In contrast, comparisons with the *r*-process element Eu show significant deviations from unity, consistent with the mixed nucleosynthetic origin and the corresponding isotopic composition. The strong correlation between Mo and Ru further supports their partially shared nucleosynthetic pathways. When studying the chemical planes of [Mo/Ru] versus [s/Mo] for the *s*-process elements Sr, Y and Zr, the resulting abundance patterns show systematic offsets with respect to pure *s*-process predictions from GCE models (Bisterzo et al. 2014), confirming that the *s*-process alone is insufficient to reproduce the observed ratios. These discrepancies cannot be attributed to NLTE effects or analysis systematics, as the abundances are either NLTE corrected or based on weak spectral lines.

While the derivation of a classical *r*-process residual for the solar system remains highly uncertain since it neglects contributions from other nucleosynthetic processes, this limitation further justifies the choice of using the *r*-II CS 22892-052 star as a representative *r*-process benchmark. Comparisons with this star and alternative nucleosynthetic scenarios further highlight the need for additional contributions beyond the classical *s*-process

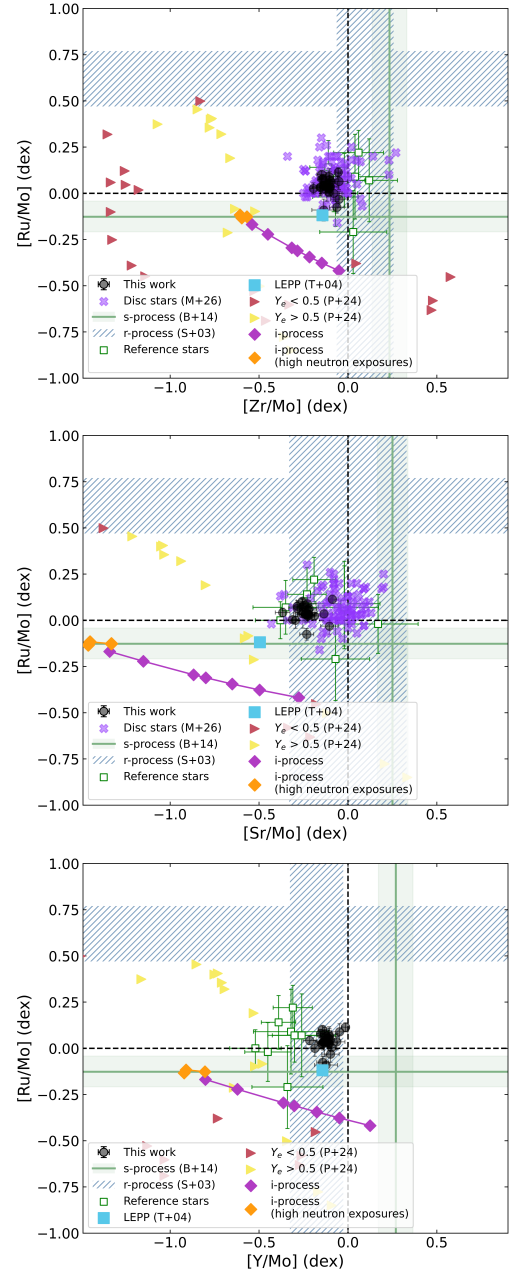


Fig. 5. [Ru/Mo] for our sample of OCs with respect to [Zr/Mo], [Sr/Mo] and [Y/Mo] based on Fig. 3 from Mishenina et al. (2026), whose results are shown as dark purple crosses. For comparison, the metal-poor stars from Honda et al. (2006, 2007), Hansen et al. (2012, 2014) and Aoki et al. (2017) are shown as white squares with green contours. The ratios from the *s*-process GCE simulations by Bisterzo et al. (2014) (green line and error range) and from the star CS 22892-052 (Snedden et al. 2003) (blue diagonal hashed lines) are included as references. We also report the LEPP prediction by Travaglio et al. 2004 (large cyan square), the ratios from different neutrino-driven ejecta conditions in Psaltis et al. (2024) with $Y_e > 0.5$ (yellow triangles, *vp*-process) and with $Y_e < 0.5$ (red triangles, weak *r*-process), along with *i*-process predictions for mild neutron exposures Mishenina et al. (2026) shown as magenta-lined rotated squares and the *i*-process predictions for high neutron exposures shown as orange rotated squares. The dashed black lines that go through [0, 0] indicate the solar value.

framework, although the relative contributions of different channels cannot be uniquely disentangled from the present data, preventing us from reliably favouring one specific combination or astrophysical source.

Overall, these results reinforce the view that Mo and Ru cannot be explained by a single dominant production site, but instead, originate from multiple nucleosynthetic processes operating on comparable timescales. As part of the future work on further constraining possible nucleosynthetic sources, Ba isotopic ratios can help distinguish between the relative contributions of the *s*-, *r*- and *i*-processes as isotopes are produced in different proportions (see Giribaldi et al. 2025, 2026, and references therein). In particular, Liu et al. (2014), showed that specific Ba isotopic signatures that could not be reproduced by standard AGB *s*-process calculations were consistent with nucleosynthesis under *i*-process conditions. In fact, the Ba isotopic ratios are becoming a target again for stars in different systems and at different metallicities (see Magrini et al. 2026). Therefore, future measurements of Ba isotopic ratios in our OC sample could provide an independent means of assessing the abundance patterns.

Data availability

Tables B.1 and C.1 are available at the CDS via <https://cdsarc.cds.unistra.fr/viz-bin/cat/J/A+A/714/A184>

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References

- Ćurjurić, B., Drazdauskas, A., Tautvaišienė, G., et al. 2026, *A&A*, 710, A86
- Alonso-Santiago, J., Frasca, A., Catanzaro, G., et al. 2021, *A&A*, 656, A149
- Alvarez, R., & Plez, B. 1998, *A&A*, 330, 1109
- Alvarez-Baena, N., Carrera, R., Thompson, H., et al. 2024, *A&A*, 687, A101
- Anders, E., & Grevesse, N. 1989, *Geochim. Cosmochim. Acta*, 53, 197
- Anguiano, B., Majewski, S. R., Hayes, C. R., et al. 2020, *AJ*, 160, 43
- Aoki, M., Ishimaru, Y., Aoki, W., & Wanaajo, S. 2017, *ApJ*, 837, 8
- Arcones, A., & Montes, F. 2011, *ApJ*, 731, 5
- Arcones, A., & Thielemann, F.-K. 2013, *J. Phys. G Nucl. Phys.*, 40, 013201
- Arnould, M., & Goriely, S. 2003, *Phys. Rep.*, 384, 1
- Asplund, M., Grevesse, N., Sauval, A. J., & Scott, P. 2009, *ARA&A*, 47, 481
- Baratella, M., D'Orazi, V., Sheminova, V., et al. 2021, *A&A*, 653, A67
- Biemont, E., Grevesse, N., Hannaford, P., & Lowe, R. M. 1981, *ApJ*, 248, 867, (BGHL)
- Bijavara Seshashayana, S., Jönsson, H., D'Orazi, V., et al. 2024, *A&A*, 683, A218
- Bijavara Seshashayana, S., Jönsson, H., D'Orazi, V., et al. 2025, *A&A*, 704, A220
- Bisterzo, S., Travaglio, C., Gallino, R., Wiescher, M., & Käppeler, F. 2014, *ApJ*, 787, 10
- Bliss, J., Arcones, A., & Qian, Y. Z. 2018, *ApJ*, 866, 105
- Bossini, D., Vallenari, A., Bragaglia, A., et al. 2019, *A&A*, 623, A108
- Bovy, J. 2015, *ApJS*, 216, 29
- Burbidge, E. M., Burbidge, G. R., Fowler, W. A., & Hoyle, F. 1957, *Rev. Mod. Phys.*, 29, 547
- Busso, M., Gallino, R., & Wasserburg, G. J. 1999, *ARA&A*, 37, 239
- Cantat-Gaudin, T., Jordi, C., Vallenari, A., et al. 2018, *A&A*, 618, A93
- Cantat-Gaudin, T., Anders, F., Castro-Ginard, A., et al. 2020, *A&A*, 640, A1
- Carbajo-Hijarrubia, J., Casamiquela, L., Carrera, R., et al. 2024, *A&A*, 687, A239
- Carrera, R., Pasquato, M., Vallenari, A., et al. 2019, *A&A*, 627, A119
- Carrera, R., Casamiquela, L., Bragaglia, A., et al. 2022, *A&A*, 663, A148
- Casali, G., Magrini, L., Frasca, A., et al. 2020, *A&A*, 643, A12
- Casamiquela, L., Carrera, R., Jordi, C., et al. 2016, *MNRAS*, 458, 3150
- Casamiquela, L., Soubiran, C., Jofré, P., et al. 2021, *A&A*, 652, A25
- Choplin, A., Siess, L., & Goriely, S. 2021, *A&A*, 648, A119
- Cirasuolo, M., Afonso, J., Carollo, M., et al. 2014, in *Ground-based and Airborne Instrumentation for Astronomy V*, 9147
- Cosentino, R., Lovis, C., Pepe, F., et al. 2012, *SPIE Conf. Ser.*, 8446, 84461V
- Côté, B., Denissenkov, P., Herwig, F., et al. 2018a, *ApJ*, 854, 105
- Côté, B., Fryer, C. L., Belczynski, K., et al. 2018b, *ApJ*, 855, 99
- Cowan, J. J., & Rose, W. K. 1977, *ApJ*, 212, 149
- Cowan, J. J., Sneden, C., Lawler, J. E., et al. 2021, *Rev. Mod. Phys.*, 93, 015002
- Dal Ponte, M., D'Orazi, V., Bragaglia, A., et al. 2025, *A&A*, 701, A289
- de Jong, R. S. 2019, *Nat. Astron.*, 3, 574
- De Silva, G. M., Freeman, K. C., Bland-Hawthorn, J., et al. 2015, *MNRAS*, 449, 2604
- D'Orazi, V., Magrini, L., Randich, S., et al. 2009, *ApJ*, 693, L31
- D'Orazi, V., Oliva, E., Bragaglia, A., et al. 2020, *A&A*, 633, A38
- D'Orazi, V., Baratella, M., Lugaro, M., Magrini, L., & Pignatari, M. 2022, *Universe*, 8, 110
- D'Orazi, V., Storm, N., Casey, A. R., et al. 2024, *MNRAS*, 531, 137
- Farouqi, K., Kratz, K.-L., Pfeiffer, B., et al. 2010, *ApJ*, 712, 1359
- Foreman-Mackey, D., Hogg, D. W., Lang, D., & Goodman, J. 2013, *PASP*, 125, 306
- Foreman-Mackey, D., Hogg, D. W., & Morton, T. D. 2014, *ApJ*, 795, 64
- Forsberg, R., Jönsson, H., Ryde, N., & Matteucci, F. 2019, *A&A*, 631, A113
- Forsberg, R., Ryde, N., Jönsson, H., Rich, R. M., & Johansen, A. 2022, *A&A*, 666, A125
- Frasca, A., Alonso-Santiago, J., Catanzaro, G., et al. 2019, *A&A*, 632, A16
- Friel, E. D. 2013, *Open Clusters and Their Role in the Galaxy* (Springer Science+Business Media Dordrecht)
- Gaia Collaboration (Prusti, T., et al.) 2016, *A&A*, 595, A1
- Gaia Collaboration (Recio-Blanco, A., et al.) 2023a, *A&A*, 674, A38
- Gaia Collaboration (Vallenari, A., et al.) 2023b, *A&A*, 674, A1
- Gallino, R., Arlandini, C., Busso, M., et al. 1998, *ApJ*, 497, 388
- Gerber, J. M., Magg, E., Plez, B., et al. 2023, *A&A*, 669, A43
- Gilmore, G., Randich, S., Asplund, M., et al. 2012, *Messenger*, 147, 25
- Giribaldi, R. E., Magrini, L., Schiappacasse-Ulloa, J., Randich, S., & Merle, T. 2025, *A&A*, 702, A65
- Giribaldi, R. E., Vescovi, D., Magrini, L., et al. 2026, *A&A*, 711, A47
- Grace, N. S., & More, K. R. 1934, *Phys. Rev.*, 45, 166
- Hansen, C. J., Primas, F., Hartman, H., et al. 2012, *A&A*, 545, A31
- Hansen, C. J., Andersen, A. C., & Christlieb, N. 2014, *A&A*, 568, A47
- Heiter, U., Lind, K., Bergemann, M., et al. 2021, *A&A*, 645, A106
- Hogg, D. W., Bovy, J., & Lang, D. 2010, arXiv e-prints [arXiv:1008.4686]

- Honda, S., Aoki, W., Ishimaru, Y., Wanajo, S., & Ryan, S. G. 2006, *ApJ*, **643**, 1180
- Honda, S., Aoki, W., Ishimaru, Y., & Wanajo, S. 2007, *ApJ*, **666**, 1189
- Jian, M., Fu, X., Matsunaga, N., et al. 2024, *A&A*, **687**, A189
- Jian, M., Fu, X., D'Orazi, V., et al. 2026, *MNRAS*, **545**, staf1797
- Jin, S., Trager, S. C., Dalton, G. B., et al. 2024, *MNRAS*, **530**, 2688
- Jofré, P., Heiter, U., Worley, C. C., et al. 2017, *A&A*, **601**, A38
- Jofré, P., Heiter, U., & Soubiran, C. 2019, *ARA&A*, **57**, 571
- Jönsson, H., Ryde, N., Nordlander, T., et al. 2017, *A&A*, **598**, A100
- Käppeler, F., Gallino, R., Bisterzo, S., & Aoki, W. 2011, *Rev. Mod. Phys.*, **83**, 157
- Karakas, A. I., & Lattanzio, J. C. 2014, *PASA*, **31**, e030
- Li, Y., & Ezzeddine, R. 2023, *AJ*, **165**, 145
- Liu, N., Savina, M. R., Davis, A. M., et al. 2014, *ApJ*, **786**, 66
- Magg, E., Bergemann, M., Serenelli, A., et al. 2022, *A&A*, **661**, A140
- Magrini, L., Spina, L., Randich, S., et al. 2018, *A&A*, **617**, A106
- Magrini, L., Viscasillas Vázquez, C., Casali, G., et al. 2022, *Universe*, **8**, 64
- Magrini, L., Viscasillas Vázquez, C., Spina, L., et al. 2023, *A&A*, **669**, A119
- Magrini, L., Bensby, T., Randich, S., et al. 2026, arXiv e-prints [arXiv:2607.04916]
- Majewski, S. R., Schiavon, R. P., Frinchaboy, P. M., et al. 2017, *AJ*, **154**, 94
- Mishenina, T., Pignatari, M., Carraro, G., et al. 2015, *MNRAS*, **446**, 3651
- Mishenina, T., Pignatari, M., Gorbaneva, T., et al. 2019, *MNRAS*, **489**, 1697
- Mishenina, T., Shereta, E., Pignatari, M., et al. 2020, *J. Phys. Stud.*, **24**, 3901
- Mishenina, T., Kurtukian-Nieto, T., Gorbaneva, T., et al. 2026, *A&A*, **705**, A38
- Montes, F., Beers, T. C., Cowan, J., et al. 2007, *ApJ*, **671**, 1685
- Murakawa, K. 1955, *J. Phys. Soc. Jpn.*, **10**, 919
- Nunnari, A., D'Orazi, V., Fiorentino, G., et al. 2026, *A&A*, **708**, A17
- Origlia, L., Dalessandro, E., Sanna, N., et al. 2019, *A&A*, **629**, A117
- Overbeek, J. C., Friel, E. D., & Jacobson, H. R. 2016, *ApJ*, **824**, 75
- Pehlivan Rhodin, A., Hartman, H., Nilsson, H., & Jönsson, P. 2024, *A&A*, **682**, A184
- Pignatari, M., Göbel, K., Reifarth, R., & Travaglio, C. 2016, *Int. J. Mod. Phys. E*, **25**, 1630003
- Plez, B. 2012, *Astrophysics Source Code Library* [record ascl:1205.004]
- Prantzos, N., Abia, C., Cristallo, S., Limongi, M., & Chieffi, A. 2020, *MNRAS*, **491**, 1832
- Psaltis, A., Jacobi, M., Montes, F., et al. 2024, *ApJ*, **966**, 11
- Randich, S., Gilmore, G., Magrini, L., et al. 2022, *A&A*, **666**, A121
- Rauscher, T., Dauphas, N., Dillmann, I., et al. 2013, *Rep. Progr. Phys.*, **76**, 066201
- Reichert, M., Obergaulinger, M., Eichler, M., Aloy, M. Á., & Arcones, A. 2021, *MNRAS*, **501**, 5733
- Roriz, M. P., Lugaro, M., Pereira, C. B., et al. 2021, *MNRAS*, **507**, 1956
- Sales-Silva, J. V., Daflon, S., Cunha, K., et al. 2022, *ApJ*, **926**, 154
- Snedden, C., Cowan, J. J., Lawler, J. E., et al. 2003, *ApJ*, **591**, 936
- Snedden, C., Cowan, J. J., & Gallino, R. 2008, *ARA&A*, **46**, 241
- Spina, L., Ting, Y.-S., De Silva, G. M., et al. 2021, *MNRAS*, **503**, 3279
- Spina, L., Magrini, L., & Cunha, K. 2022, *Universe*, **8**, 87
- Spite, F., Spite, M., Barbuy, B., et al. 2018, *A&A*, **611**, A30
- Storm, N., & Bergemann, M. 2023, *MNRAS*, **525**, 3718
- Tarricq, Y., Soubiran, C., Casamiquela, L., et al. 2021, *A&A*, **647**, A19
- Taylor, M. B. 2005, *ASP Conf. Ser.*, **347**, 29
- Travaglio, C., Gallino, R., Arnone, E., et al. 2004, *ApJ*, **601**, 864
- van der Plas, G., van den Ancker, M. E., Fedele, D., et al. 2008, *A&A*, **485**, 487
- Van der Swaelmen, M., Viscasillas Vázquez, C., Cescutti, G., et al. 2023, *A&A*, **670**, A129
- Velichko, A. B., Mashonkina, L. I., & Nilsson, H. 2010, *Astron. Lett.*, **36**, 664
- Wanajo, S., Janka, H.-T., & Müller, B. 2011, *ApJ*, **726**, L15
- Whaling, W., & Brault, J. W. 1988, *Phys. Scr.*, **38**, 707, (WBb)
- Wickliffe, M. E., Salih, S., & Lawler, J. E. 1994, *J. Quant. Spectr. Rad. Transf.*, **51**, 545, (WSL)
- Zhang, R., Lucatello, S., Bragaglia, A., et al. 2022, *A&A*, **667**, A103

Appendix A: Colour-magnitude diagrams (CMDs) for the studied clusters

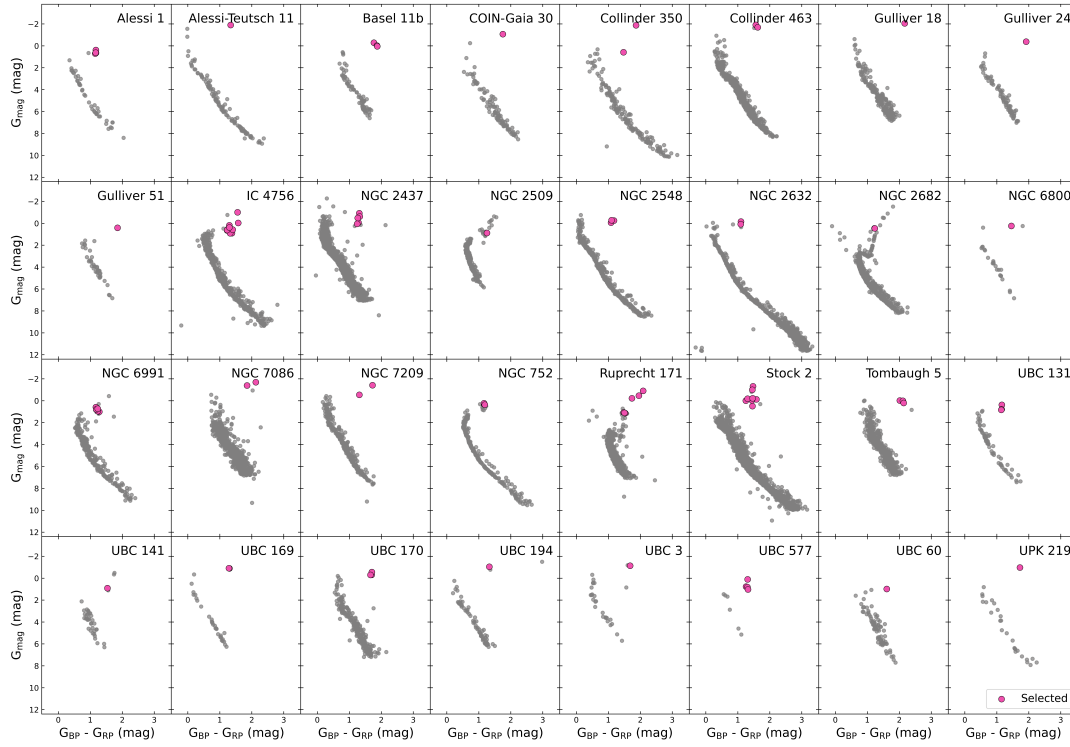


Fig. A.1. CMDs of the studied OCs using the members determined by [Cantat-Gaudin et al. \(2020\)](#).

Appendix B: Stellar parameters for the studied members

Table B.1. Stellar parameters for 81 stars considered in this work.

Star	Gaia DR3 ID	T_{eff}^a (K)	$\log g^a$ (dex)	v_{mic}^a kms^{-1}	v_{mac}^b kms^{-1}	$[\text{Fe}/\text{H}]^a$ (dex)
Alessi 161 star #1	4505874693061489280	4731 ± 32	2.32 ± 0.09	1.84 ± 0.01	4.16 ± 0.21	0.00 ± 0.01
Alessi 1 star #2	402506369136008832	4986 ± 30	2.83 ± 0.07	1.30 ± 0.01	3.22 ± 0.06	-0.00 ± 0.01
Alessi 1 star #3	402505991180022528	4996 ± 30	2.84 ± 0.07	1.28 ± 0.01	3.11 ± 0.08	-0.01 ± 0.01
Alessi 1 star #5	402867593065772288	4939 ± 31	2.73 ± 0.08	1.35 ± 0.01	3.29 ± 0.07	-0.03 ± 0.01
Alessi 1 star #6	402880684126058880	4985 ± 32	2.87 ± 0.08	1.28 ± 0.01	2.85 ± 0.03	-0.01 ± 0.01

Notes. ^aObtained by [DP25](#). ^bEstimated for this work using the results from [DP25](#). Only an portion of the table is shown here. The full table is available at the CDS.

Appendix C: Estimated abundances star by star

Table C.1. Estimated abundances for Mo, Ru and Zr for each star in our sample, together with the number of spectral lines used to derive the mean abundances of Mo and Zr.

Star	$[\text{Mo}/\text{Fe}]$ (dex)	$N_{\text{lines}}^{\text{Mo}}$	$[\text{Ru}/\text{Fe}]$ (dex)	$[\text{Zr}/\text{Fe}]$ (dex)	$N_{\text{lines}}^{\text{Zr}}$
Alessi 161 star #1	0.06 ± 0.01	2	0.13 ± 0.04	-0.04 ± 0.03	2
Alessi 1 star #2	0.06 ± 0.04	1	0.16 ± 0.05	-0.06 ± 0.01	2
Alessi 1 star #3	0.14 ± 0.04	1	0.04 ± 0.05	-0.03 ± 0.02	2
Alessi 1 star #5	0.06 ± 0.01	2	0.09 ± 0.05	-0.08 ± 0.02	2
Alessi 1 star #6	0.05 ± 0.04	1	0.11 ± 0.05	-0.08 ± 0.01	2

Notes. The number of lines for Ru is not reported, as it is based on a single line. Only an portion of the table is shown here. The full table is available at the CDS.

Appendix D: Abundance sensitivities**Table D.1.** Mo abundance sensitivities to change in stellar parameters.

Star	ΔT_{eff} (+100 K)	$\Delta \log g$ (+0.25 dex)	Δv_{mic} (+0.5 km s ⁻¹)	$\Delta [\text{Fe}/\text{H}]$ (+0.25 dex)	$\Delta [\text{C}/\text{Fe}]$ (+0.25 dex)	$\Delta [\text{N}/\text{Fe}]$ (+0.25 dex)
Alessi 1 star #2	0.14	0.00	0.00	-0.29	-0.07	0.00
Alessi 1 star #3	0.13	0.01	-0.01	-0.28	-0.06	0.00
Alessi 1 star #5	0.14	0.01	0.00	-0.32	-0.16	-0.08
Alessi 1 star #6	0.14	0.00	0.00	-0.28	-0.05	0.00
Alessi 161 star #1	0.15	0.02	0.00	-0.31	-0.15	-0.07
Alessiteutsch 11 star #1	0.14	0.03	-0.01	-0.28	-0.10	-0.03
Basel 11 star #1	0.14	0.01	0.00	-0.33	-0.14	0.00
Basel 11 star #2	0.14	0.01	0.00	-0.32	-0.12	0.00
Basel 11 star #3	0.14	0.02	-0.01	-0.28	-0.09	0.01
COIN-Gaia30 star #1	0.14	0.01	-0.01	-0.30	-0.10	0.00
Collinder 350 star #2	0.13	0.00	0.00	-0.38	-0.15	-0.01
Collinder 463 star #1	0.15	0.02	-0.01	-0.29	-0.15	-0.05
Collinder 463 star #3	0.15	0.02	0.00	-0.30	-0.14	-0.05
Gulliver 51 star #1	0.14	0.02	0.00	-0.30	-0.16	-0.07
IC 4756 star #0	0.13	0.01	0.00	-0.30	-0.08	0.00
IC 4756 star #2	0.14	0.03	-0.02	-0.27	-0.08	-0.03
IC 4756 star #4	0.15	0.01	-0.03	-0.26	–	0.00
IC 4756 star #5	0.14	0.00	0.01	-0.37	-0.28	-0.07
IC 4756 star #7	0.13	0.00	0.00	-0.29	-0.06	0.00
IC 4756 star #8	0.14	0.01	0.00	-0.33	-0.26	-0.08
IC 4756 star #9	0.13	-0.01	0.00	-0.30	-0.06	0.00
IC 4756 star #11	0.14	0.01	0.00	-0.32	-0.14	-0.09
IC 4756 star #12	0.13	0.00	-0.01	-0.30	-0.07	0.00
IC 4756 star #13	0.13	-0.01	0.00	-0.31	–	0.00
IC 4756 star #14	0.13	-0.01	0.00	-0.28	–	0.00
IC 4756 star #15	0.13	0.00	0.00	-0.29	–	0.00
LP 1800 star #1	0.15	0.02	-0.01	-0.36	-0.13	-0.12
LP 1800 star #2	0.14	0.01	-0.01	-0.36	-0.18	-0.01
LP 1800 star #3	0.15	0.01	0.00	-0.35	-0.23	-0.11
NGC 2437 star #1	0.15	0.01	0.00	-0.31	-0.15	-0.06
NGC 2437 star #3	0.15	0.02	-0.01	-0.43	-0.16	0.00
NGC 2437 star #4	0.15	0.01	0.01	-0.30	-0.12	-0.06
NGC 2437 star #5	0.14	0.00	-0.01	-0.27	-0.04	0.00
NGC 2437 star #10	0.16	0.03	-0.02	-0.29	-0.15	-0.05
NGC 2509 star #18	0.14	0.01	-0.03	-0.27	-0.08	0.00
NGC 2548 star #1	0.13	0.00	-0.01	-0.42	-0.25	-0.01
NGC 2548 star #2	0.14	0.01	0.00	-0.32	-0.13	-0.06
NGC 2548 star #3	0.14	0.02	0.00	-0.30	-0.15	-0.07
NGC 2632 star #1	0.14	0.03	0.00	-0.34	-0.29	-0.08
NGC 2632 HD73665	0.15	0.02	0.01	-0.33	-0.27	-0.08
NGC 2682 star #8	0.14	0.02	-0.01	-0.29	-0.12	-0.05
NGC 2682 star #9	0.14	0.02	-0.01	-0.30	-0.15	-0.08
NGC 6800 star #1	0.13	0.01	-0.01	-0.29	-0.07	0.00
NGC 6991 star #4	0.13	0.00	0.00	-0.28	-0.05	0.00
NGC 6991 star #6	0.20	0.00	-0.01	-0.32	-0.16	-0.08
NGC 6991 star #7	0.13	0.00	-0.01	-0.27	-0.04	0.00
NGC 6991 star #9	0.13	0.00	0.00	-0.27	-0.04	0.00
NGC 6991 star #10	0.14	0.00	0.01	-0.32	-0.14	0.06
NGC 7086 star #1	0.15	0.03	-0.01	-0.29	-0.12	-0.05
NGC 7086 star #2	0.15	0.03	-0.01	-0.28	-0.12	-0.05
NGC 7209 star #2	0.16	0.02	0.00	-0.36	-0.50	-0.12
NGC 752 star #3	0.14	0.02	0.04	-0.33	-0.22	-0.13
NGC 752 star #4	0.14	0.02	0.01	-0.31	-0.18	-0.11
NGC 752 star #5	0.14	0.02	0.02	-0.32	-0.19	-0.11

Table D.1. Continued.

Star	ΔT_{eff} (+100 K)	$\Delta \log g$ (+0.25 dex)	Δv_{mic} (+0.5 km s ⁻¹)	$\Delta [\text{Fe}/\text{H}]$ (+0.25 dex)	$\Delta [\text{C}/\text{Fe}]$ (+0.25 dex)	$\Delta [\text{N}/\text{Fe}]$ (+0.25 dex)
Ruprecht 171 star #4	0.07	-0.06	-0.08	-0.34	-0.12	-0.08
Ruprecht 171 star #5	0.14	0.01	-0.01	-0.27	-0.06	-0.01
Ruprecht 171 star #6	0.07	-0.07	-0.08	-0.34	-0.12	-0.08
Ruprecht 171 star #8	0.14	0.01	-0.02	-0.26	-0.04	0.00
Stock 2 star #1	0.15	0.01	0.00	-0.33	-0.27	-0.08
Stock 2 star #2	0.14	0.03	-0.02	-0.23	-0.05	-0.03
Stock 2 star #4	0.15	0.01	0.01	-0.35	-0.23	-0.07
Stock 2 star #6	0.15	0.02	0.00	-0.36	—	-0.09
Stock 2 star #7	0.15	0.02	0.01	-0.41	-0.38	-0.10
Stock 2 star #8	0.15	0.01	0.00	-0.37	-0.26	-0.12
Stock 2 star #9	0.14	0.01	0.00	-0.35	-0.24	-0.07
Stock 2 star #10	0.14	0.02	-0.01	-0.31	-0.13	-0.05
Tombaugh 5 star #2	0.14	0.01	0.00	-0.28	-0.08	0.00
Tombaugh 5 star #3	0.14	-0.02	0.01	-0.32	-0.12	0.01
Tombaugh 5 star #4	0.15	0.02	0.00	-0.32	-0.14	0.00
UBC 141 star #1	0.14	0.01	-0.02	-0.26	-0.03	0.00
UBC 169 star #2	0.14	0.01	-0.01	-0.35	-0.13	-0.01
UBC 194 star #1	0.14	0.02	0.00	-0.41	-0.20	-0.01
UBC 577 star #1	—	—	—	—	—	—
UBC 577 star #2	0.13	0.00	0.00	-0.31	0.02	0.00
UBC 577 star #3	0.13	0.00	0.00	-0.30	-0.07	0.00
UBC 577 star #5	0.13	-0.01	0.00	-0.28	-0.06	0.00
UBC 60 star #3	0.14	0.01	-0.01	-0.31	-0.26	-0.07
UPK 219 star #1	0.19	0.01	0.02	-0.31	-0.16	-0.07
UPK 84 star #1	0.13	0.00	0.00	-0.29	-0.05	0.00
UPK 84 star #2	0.13	0.00	0.00	-0.28	-0.04	0.00
UPK 84 star #3	0.15	0.01	0.01	-0.37	-0.24	-0.13

Notes. Some sensitivities are not reported either because there was no [Mo/Fe] estimate or because the resulting fit for the test parameters was unsuccessful.

Table D.2. Ru abundance sensitivities to change in stellar parameters.

Star	ΔT_{eff} (+100 K)	$\Delta \log g$ (+0.25 dex)	Δv_{mic} (+0.5 km s ⁻¹)	$\Delta [\text{Fe}/\text{H}]$ (+0.25 dex)	$\Delta [\text{C}/\text{Fe}]$ (+0.25 dex)	$\Delta [\text{N}/\text{Fe}]$ (+0.25 dex)
Alessi 1 star #2	0.02	0.02	0.01	-0.30	-0.04	-0.01
Alessi 1 star #3	0.04	0.01	0.03	-0.30	-0.03	0.01
Alessi 1 star #5	0.01	0.01	0.00	-0.31	-0.04	-0.01
Alessi 1 star #6	0.02	0.01	0.01	-0.30	-0.05	-0.01
Alessi 161 star #1	0.02	0.02	0.00	-0.26	-0.03	-0.01
Alessiteutsch 11 star #1	0.04	0.04	-0.01	-0.24	-0.03	-0.01
Basel 11 star #1	—	—	—	—	—	—
Basel 11 star #2	—	—	—	—	—	—
Basel 11 star #3	—	—	—	—	—	—
COIN-Gaia30 star #1	0.02	0.02	0.00	-0.29	-0.05	-0.01
Collinder 350 star #2	—	—	—	—	—	—
Collinder 463 star #1	0.03	0.03	0.00	-0.24	-0.03	-0.01
Collinder 463 star #3	0.03	0.03	0.00	-0.25	-0.03	-0.01
Gulliver 51 star #1	0.03	0.03	0.00	-0.25	-0.04	-0.01
IC 4756 star #0	0.01	0.01	0.01	-0.36	-0.07	-0.01
IC 4756 star #2	0.04	0.04	-0.01	-0.23	-0.02	-0.01
IC 4756 star #4	0.04	0.04	-0.01	-0.24	-0.03	-0.01
IC 4756 star #5	0.00	0.00	0.01	-0.31	-0.04	-0.01
IC 4756 star #7	0.01	0.01	0.01	-0.32	-0.05	-0.01
IC 4756 star #8	0.01	0.01	0.01	-0.33	-0.04	-0.01

Table D.2. Continued.

Star	ΔT_{eff} (+100 K)	$\Delta \log g$ (+0.25 dex)	Δv_{mic} (+0.5 km s ⁻¹)	$\Delta [\text{Fe}/\text{H}]$ (+0.25 dex)	$\Delta [\text{C}/\text{Fe}]$ (+0.25 dex)	$\Delta [\text{N}/\text{Fe}]$ (+0.25 dex)
IC 4756 star #9	—	—	—	—	—	—
IC 4756 star #11	0.01	0.01	0.01	-0.31	-0.04	-0.01
IC 4756 star #12	0.01	0.01	0.00	-0.33	-0.05	-0.01
IC 4756 star #13	0.02	0.01	0.02	-0.37	-0.04	0.00
IC 4756 star #14	0.00	0.00	0.01	-0.34	-0.02	-0.01
IC 4756 star #15	0.02	0.02	0.00	-0.30	-0.04	-0.01
LP 1800 star #1	0.02	0.02	0.00	-0.29	-0.05	-0.02
LP 1800 star #2	—	—	—	—	—	—
LP 1800 star #3	0.01	0.01	0.01	-0.31	-0.05	-0.02
NGC 2437 star #1	-0.03	0.02	0.00	-0.31	-0.04	-0.01
NGC 2437 star #3	—	—	—	—	—	—
NGC 2437 star #4	0.02	0.02	0.00	-0.27	-0.04	-0.01
NGC 2437 star #5	-0.01	0.00	-0.03	-0.32	-0.06	-0.03
NGC 2437 star #10	0.04	0.04	-0.02	-0.25	-0.03	-0.01
NGC 2509 star #18	—	—	—	—	—	—
NGC 2548 star #1	—	—	—	—	—	—
NGC 2548 star #2	0.00	0.04	-0.01	-0.33	-0.03	-0.03
NGC 2548 star #3	0.03	0.03	0.01	-0.25	-0.03	0.00
NGC 2632 star #1	0.02	0.02	0.00	-0.28	-0.03	-0.01
NGC 2632 HD73665	0.02	0.02	0.01	-0.27	-0.03	-0.02
NGC 2682 star #8	0.03	0.03	-0.02	-0.25	-0.03	-0.01
NGC 2682 star #9	0.03	0.03	0.00	-0.25	-0.03	-0.01
NGC 6800 star #1	0.02	0.02	0.00	-0.30	-0.04	-0.01
NGC 6991 star #4	0.00	0.00	0.00	-0.31	-0.03	0.00
NGC 6991 star #6	0.02	0.02	0.00	-0.28	-0.04	-0.01
NGC 6991 star #7	0.01	0.01	0.01	-0.31	-0.04	-0.01
NGC 6991 star #9	0.01	0.01	0.01	-0.30	-0.03	-0.01
NGC 6991 star #10	0.00	0.00	0.01	-0.31	-0.04	-0.01
NGC 7086 star #1	0.03	0.03	0.00	-0.23	-0.03	-0.01
NGC 7086 star #2	0.04	0.05	-0.01	-0.24	-0.02	-0.02
NGC 7209 star #2	0.01	0.01	0.00	-0.30	-0.05	-0.01
NGC 752 star #3	0.02	0.02	0.00	-0.28	-0.03	-0.01
NGC 752 star #4	0.02	0.02	0.00	-0.28	-0.03	-0.01
NGC 752 star #5	0.02	0.02	0.00	-0.28	-0.03	-0.01
Ruprecht 171 star #4	0.03	0.03	0.00	-0.25	-0.04	-0.01
Ruprecht 171 star #5	0.03	0.03	0.00	-0.25	-0.04	-0.01
Ruprecht 171 star #6	0.04	0.03	0.01	-0.24	-0.04	0.00
Ruprecht 171 star #8	0.03	0.03	0.00	-0.24	-0.04	-0.01
Stock 2 star #1	0.01	0.01	0.01	-0.30	-0.04	-0.01
Stock 2 star #2	0.04	0.04	-0.01	-0.23	-0.02	0.00
Stock 2 star #4	0.01	0.01	0.01	-0.30	-0.04	-0.01
Stock 2 star #6	—	—	—	—	—	—
Stock 2 star #7	0.01	0.01	0.00	-0.28	-0.04	-0.01
Stock 2 star #8	—	—	—	—	—	—
Stock 2 star #9	0.02	0.02	0.00	-0.34	-0.05	-0.01
Stock 2 star #10	0.02	0.02	-0.01	-0.28	-0.04	-0.01
Tombaugh 5 star #2	—	—	—	—	—	—
Tombaugh 5 star #3	—	—	—	—	—	—
Tombaugh 5 star #4	0.03	0.03	0.00	-0.28	-0.04	-0.01
UBC 141 star #1	0.04	0.03	-0.01	-0.26	-0.03	-0.01
UBC 169 star #2	—	—	—	—	—	—
UBC 194 star #1	—	—	—	—	—	—
UBC 577 star #1	-0.01	-0.01	0.02	-0.37	-0.06	-0.01
UBC 577 star #2	0.00	0.00	0.01	-0.34	-0.14	-0.01
UBC 577 star #3	0.01	0.01	0.01	-0.38	-0.09	-0.02
UBC 577 star #5	—	—	—	—	—	—
UBC 60 star #3	0.02	0.02	0.00	-0.29	-0.03	-0.01

Notes. Some sensitivities are not reported either because there was no [Ru/Fe] estimate or because the resulting fit for the test parameters was unsuccessful.

Table D.3. Zr abundance sensitivities to change in stellar parameters.

Star	ΔT_{eff} (+100 K)	$\Delta \log g$ (+0.25 dex)	Δv_{mic} (+0.5 km s ⁻¹)	$\Delta [\text{Fe}/\text{H}]$ (+0.25 dex)	$\Delta [\text{C}/\text{Fe}]$ (+0.25 dex)	$\Delta [\text{N}/\text{Fe}]$ (+0.25 dex)
UPK 219 star #1	0.01	0.01	0.00	-0.30	-0.04	-0.01
UPK 84 star #1	0.00	0.00	0.00	-0.31	-0.04	-0.01
UPK 84 star #2	0.00	0.00	0.01	-0.33	-0.05	-0.01
UPK 84 star #3	0.00	0.00	0.00	-0.32	-0.04	-0.01
Alessi 1 star #2	0.16	0.01	0.00	-0.29	-0.02	-0.01
Alessi 1 star #3	0.16	0.01	0.00	-0.29	-0.02	-0.01
Alessi 1 star #5	0.17	0.00	0.00	-0.29	-0.02	-0.01
Alessi 1 star #6	0.17	0.01	0.00	-0.29	-0.01	-0.01
Alessi 161 star #1	0.19	0.01	-0.01	-0.28	-0.01	-0.01
Alessiteutsch 11 star #1	0.20	0.02	-0.03	-0.27	-0.01	-0.01
Basel 11 star #1	0.17	0.00	0.00	-0.29	-0.02	-0.01
Basel 11 star #2	0.16	0.00	0.00	-0.28	-0.01	-0.01
Basel 11 star #3	0.19	0.01	-0.02	-0.28	-0.01	-0.01
COIN-Gaia30 star #1	0.18	0.01	-0.01	-0.28	-0.01	-0.01
Collinder 350 star #2	0.15	0.00	0.00	-0.31	-0.02	-0.02
Collinder 463 star #1	0.20	0.02	-0.02	-0.27	-0.01	-0.01
Collinder 463 star #3	0.20	0.01	-0.01	-0.28	-0.01	-0.01
Gulliver 51 star #1	0.20	0.01	-0.01	-0.27	-0.01	-0.01
IC 4756 star #0	0.16	0.01	0.00	-0.29	-0.01	-0.01
IC 4756 star #2	0.20	0.03	-0.06	-0.27	-0.01	0.00
IC 4756 star #4	0.20	0.02	-0.04	-0.27	-0.01	-0.01
IC 4756 star #5	0.16	0.00	0.00	-0.28	-0.01	-0.01
IC 4756 star #7	0.15	0.00	-0.01	-0.30	-0.03	-0.03
IC 4756 star #8	0.16	0.01	0.01	-0.28	-0.01	-0.01
IC 4756 star #9	0.15	0.00	0.01	-0.29	-0.02	-0.01
IC 4756 star #11	0.16	0.01	0.00	-0.29	-0.01	-0.01
IC 4756 star #12	0.16	0.01	0.00	-0.29	-0.01	-0.01
IC 4756 star #13	0.15	0.00	0.01	-0.29	-0.01	-0.01
IC 4756 star #14	0.15	0.00	0.01	-0.28	-0.01	-0.01
IC 4756 star #15	0.17	0.01	0.00	-0.28	-0.01	-0.01
LP 1800 star #1	0.18	0.00	-0.01	-0.30	-0.02	-0.02
LP 1800 star #2	0.17	0.01	-0.01	-0.31	-0.03	-0.02
LP 1800 star #3	0.18	0.00	0.00	-0.29	-0.02	-0.02
NGC 2437 star #1	0.17	0.00	-0.01	-0.29	-0.01	-0.01
NGC 2437 star #3	0.17	0.01	0.01	-0.29	-0.01	0.00
NGC 2437 star #4	0.18	0.00	0.00	-0.28	-0.01	-0.01
NGC 2437 star #5	0.17	0.00	0.00	-0.28	-0.01	-0.01
NGC 2437 star #10	0.21	0.03	-0.05	-0.28	-0.01	-0.01
NGC 2509 star #18	0.18	0.02	-0.03	-0.29	-0.01	0.00
NGC 2548 star #1	0.16	0.00	0.00	-0.31	-0.02	-0.01
NGC 2548 star #2	0.16	0.01	0.00	-0.29	-0.01	-0.01
NGC 2548 star #3	0.19	0.01	-0.01	-0.28	-0.01	-0.01
NGC 2632 star #1	0.17	0.01	-0.01	-0.28	-0.02	-0.01
NGC 2632 HD73665	0.18	0.01	0.00	-0.29	-0.02	-0.01
NGC 2682 star #8	0.19	0.01	-0.02	-0.28	-0.01	-0.01
NGC 2682 star #9	0.19	0.01	-0.02	-0.28	-0.01	-0.01
NGC 6800 star #1	0.16	0.01	-0.01	-0.28	-0.01	-0.01
NGC 6991 star #4	0.16	0.00	0.00	-0.28	-0.01	-0.01
NGC 6991 star #6	0.18	0.01	-0.01	-0.28	-0.01	-0.01
NGC 6991 star #7	0.17	0.01	0.00	-0.28	-0.01	-0.01
NGC 6991 star #9	0.16	0.00	0.01	-0.28	-0.01	-0.01
NGC 6991 star #10	0.16	0.00	0.00	-0.28	-0.01	-0.01
NGC 7086 star #1	0.20	0.02	-0.02	-0.27	-0.01	-0.01

Table D.3. Continued.

Star	ΔT_{eff} (+100 K)	$\Delta \log g$ (+0.25 dex)	Δv_{mic} (+0.5 km s ⁻¹)	$\Delta [\text{Fe}/\text{H}]$ (+0.25 dex)	$\Delta [\text{C}/\text{Fe}]$ (+0.25 dex)	$\Delta [\text{N}/\text{Fe}]$ (+0.25 dex)
NGC 7086 star #2	0.20	0.03	-0.03	-0.27	-0.01	0.00
NGC 7209 star #2	0.18	0.01	0.00	-0.29	-0.02	-0.01
NGC 752 star #3	0.17	0.01	-0.01	-0.28	-0.01	-0.01
NGC 752 star #4	0.17	0.01	-0.01	-0.28	-0.01	-0.01
NGC 752 star #5	0.17	0.01	-0.01	-0.28	-0.01	-0.01
Ruprecht 171 star #4	0.19	0.01	-0.01	-0.28	-0.02	-0.01
Ruprecht 171 star #5	0.19	0.01	-0.01	-0.28	-0.01	-0.01
Ruprecht 171 star #6	0.19	0.01	-0.01	-0.28	-0.02	-0.01
Ruprecht 171 star #8	0.19	0.01	-0.02	-0.27	-0.01	-0.01
Stock 2 star #1	0.18	0.00	0.00	-0.29	-0.02	-0.01
Stock 2 star #2	0.19	0.03	-0.05	-0.27	-0.01	0.00
Stock 2 star #4	0.17	0.00	0.00	-0.28	-0.01	-0.01
Stock 2 star #6	0.16	0.01	0.00	-0.29	-0.01	-0.01
Stock 2 star #7	0.18	0.01	0.00	-0.28	-0.01	-0.01
Stock 2 star #8	0.16	0.01	0.00	-0.30	-0.02	-0.01
Stock 2 star #9	0.16	0.01	-0.01	-0.29	-0.01	-0.01
Stock 2 star #10	0.18	0.01	-0.03	-0.28	-0.01	-0.01
Tombaugh 5 star #2	0.18	0.01	0.00	-0.28	-0.01	-0.01
Tombaugh 5 star #3	0.18	0.01	0.01	-0.28	-0.01	-0.01
Tombaugh 5 star #4	0.19	0.01	-0.01	-0.29	-0.02	-0.01
UBC 141 star #1	0.19	0.01	-0.03	-0.27	-0.01	-0.01
UBC 169 star #2	0.17	0.00	-0.01	-0.30	-0.02	-0.01
UBC 194 star #1	0.16	0.03	-0.01	-0.30	-0.02	-0.01
UBC 577 star #1	0.15	0.00	0.01	-0.28	-0.01	-0.01
UBC 577 star #2	0.16	0.00	0.01	-0.28	-0.01	-0.01
UBC 577 star #3	0.16	0.01	0.00	-0.29	-0.02	-0.01
UBC 577 star #5	0.15	0.00	0.01	-0.29	-0.01	-0.01
UBC 60 star #3	0.17	0.01	-0.01	-0.29	-0.01	-0.01
UPK 219 star #1	0.18	0.01	-0.01	-0.29	-0.01	-0.01
UPK 84 star #1	0.16	0.00	0.01	-0.29	-0.02	-0.02
UPK 84 star #2	0.16	0.00	0.00	-0.29	-0.02	-0.01
UPK 84 star #3	0.15	0.00	0.00	-0.29	-0.02	-0.02